

QCon
全球软件开发大会

自回归模型 在视觉生成领域的应用

韩剑

字节跳动 / 商业化GenAI算法工程师



极客邦科技 2026 年会议规划

促进软件开发及相关领域知识与创新的传播



参会咨询

🕒 6月 📍 上海

👥 1000人

AiCon

全球人工智能开发与应用大会

会议时间：6月26-27日

- AI Infra 系统工程
- 多 Agent 协作与实践
- 多模态融合
- 模型训练与推理创新
- 数据平台与特征服务

🕒 8月 📍 深圳

👥 1000人

AiCon

全球人工智能开发与应用大会

会议时间：8月21-22日

- Agentic AI
- 轻量化与高效推理
- 多模态应用
- AI + IoT 场景实践
- AI 工业化落地

🕒 10月 📍 上海

👥 1200人

QCon

全球软件开发大会

会议时间：10月22-24日

- AI Agent
- Vibe Coding
- 智能可观测
- 推理基建
- 模型攻防
- AI x 创造力

🕒 12月 📍 北京

👥 1000人

AiCon

全球人工智能开发与应用大会

会议时间：12月18-19日

- 大模型架构创新
- 多模态 AI 产业融合
- 具身智能
- AI for Science
- 大模型安全

目录

01

自回归模型 vs. 扩散模型

02

自回归文生图模型: Infinity

03

自回归视频生成模型: InfinityStar

04

生成式精炼网络: GRN

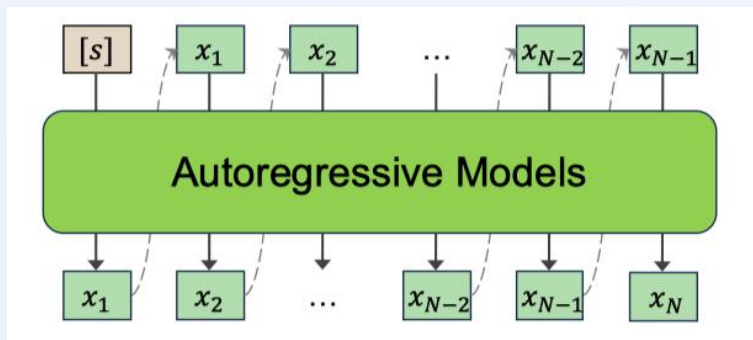
05

核心要点

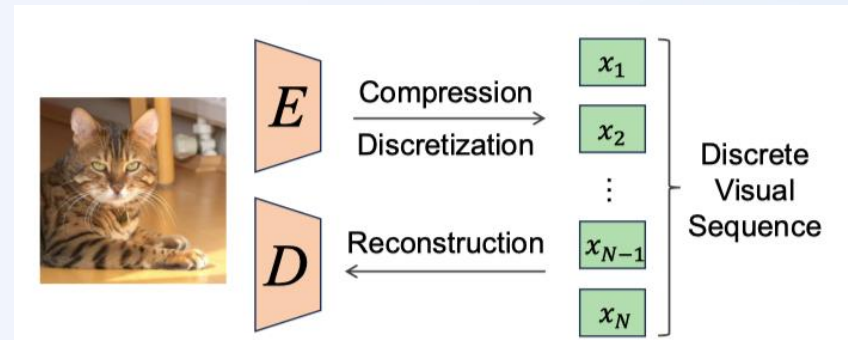
01

自回归模型 vs. 扩散模型

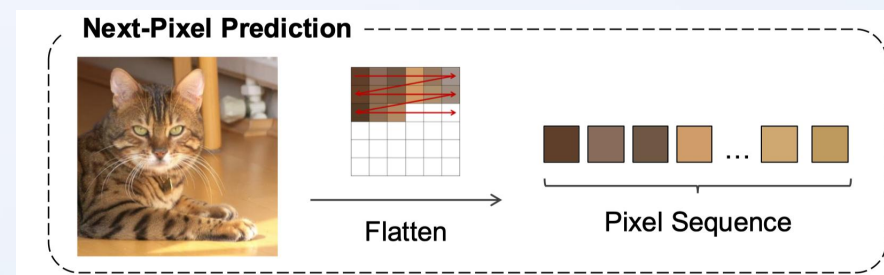
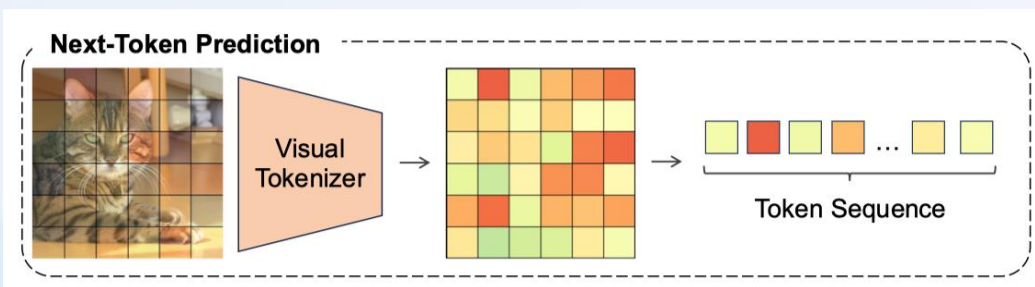
AutoRegressive Models



Autoregressive Sequence Modeling

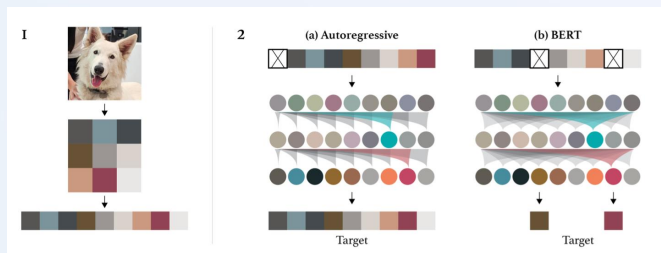


Sequence Representation

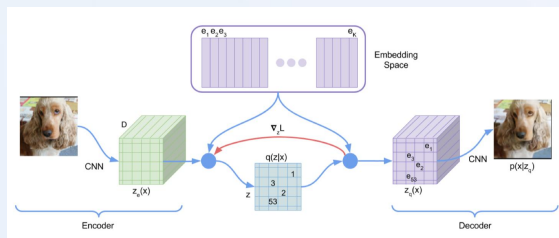


credit: Autoregressive Models in Vision: A Survey

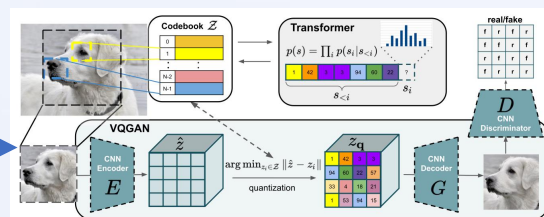
AutoRegressive Models



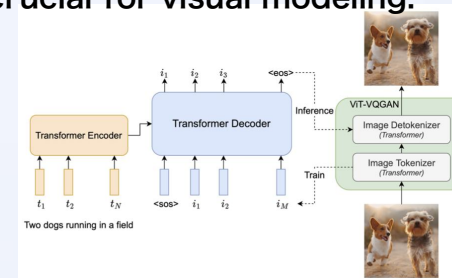
[2020] iGPT: Generative Image & Pretraining from Pixels



[2017] VQVAE: Tokenize Images into discrete token index



[2021] VQGAN: Image tokenizer + AutoRegressive transformer



[2022] Parti : Scaling Up the AutoRegressive transformer

Challenges:

- Autoregressive models perform significantly worse than sota diffusion models in high-resolution image synthesis.
- Autoregressive models have not demonstrated the same scaling law properties as LLMs in text generation.
- Due to raster order prediction, autoregressive models suffer from very slow prediction speeds.
- The raster-scan order is not the most natural "order" for images, as it loses the global information crucial for visual modeling.

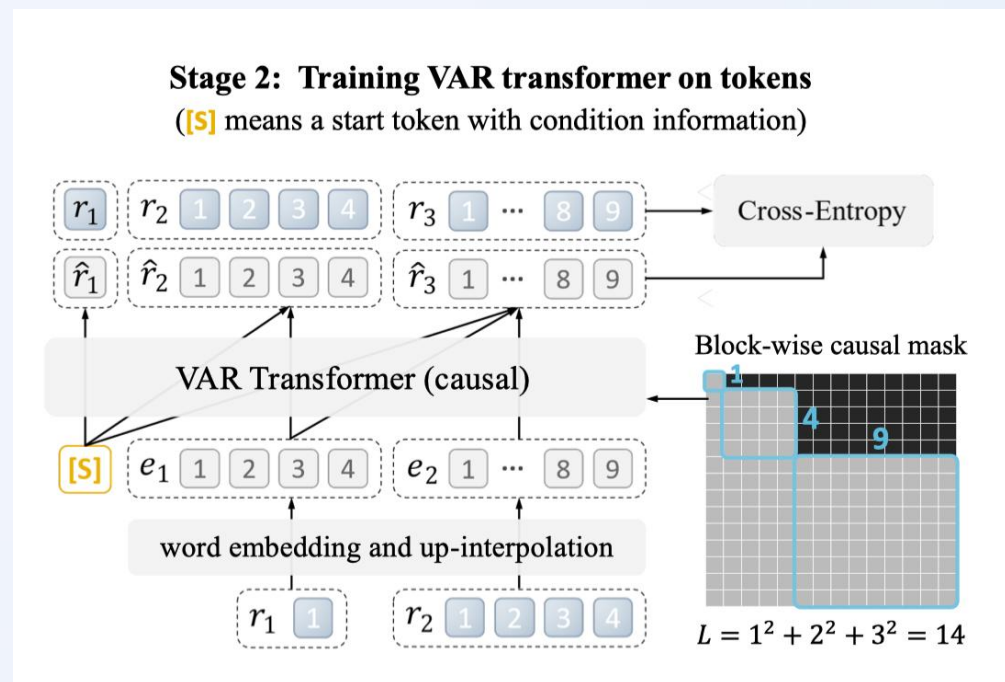
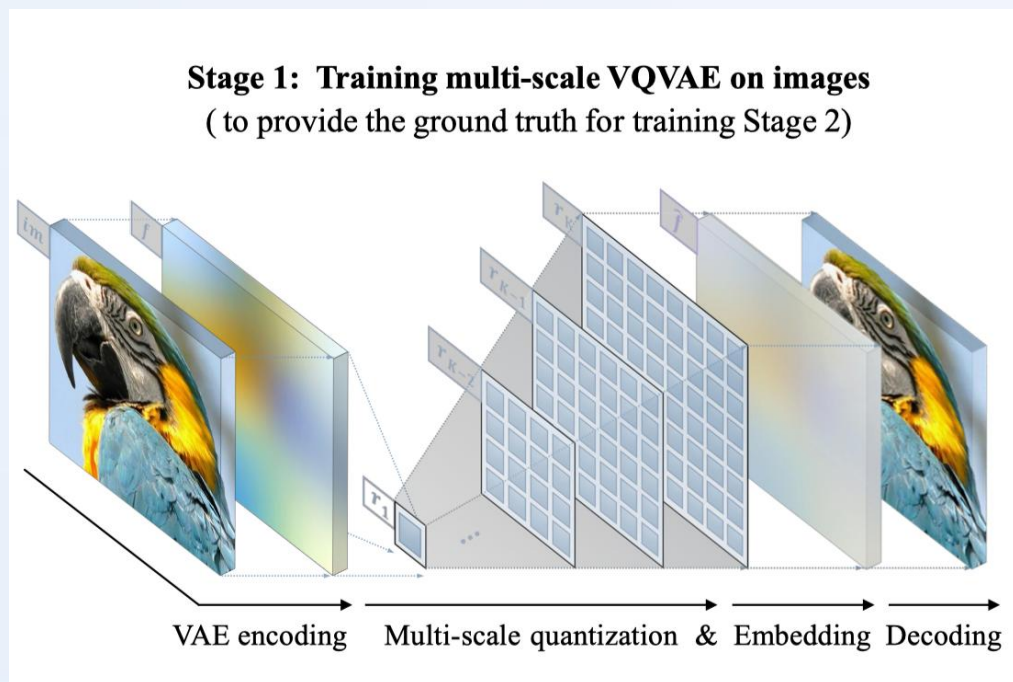
[1] Mark Chen et al., Generative Pretraining From Pixels, in ICML, 2020.

[2] Aaron van et al., Neural Discrete Representation Learning, 2017.

[3] PATRICK ESSER et al., Taming Transformers for High-Resolution Image Synthesis, 2021 CVPR.

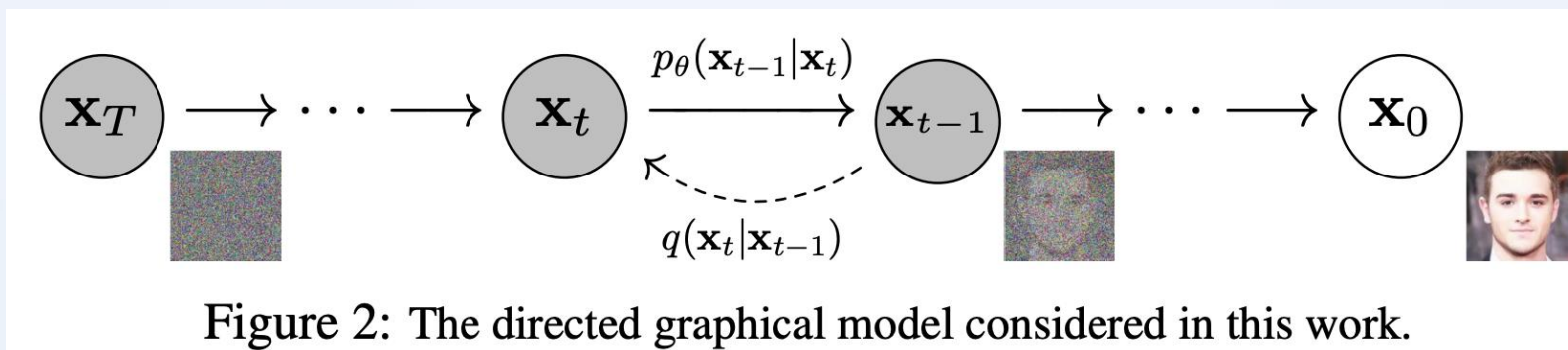
[4] Jiahui Yu et al., Scaling Autoregressive Models for Content-Rich Text-to-Image Generation, 2022 Arxiv.

Visual Autoregressive Models



- Stage1 : Train multi-scale Image Tokenization for Images, which Tokenize Images into discrete token index
- Stage2: Train **GPT-style (AutoRegressive in scale space)** transformers on VQ-Tokens with teacher forcing

Diffusion Models



Algorithm 1 Training

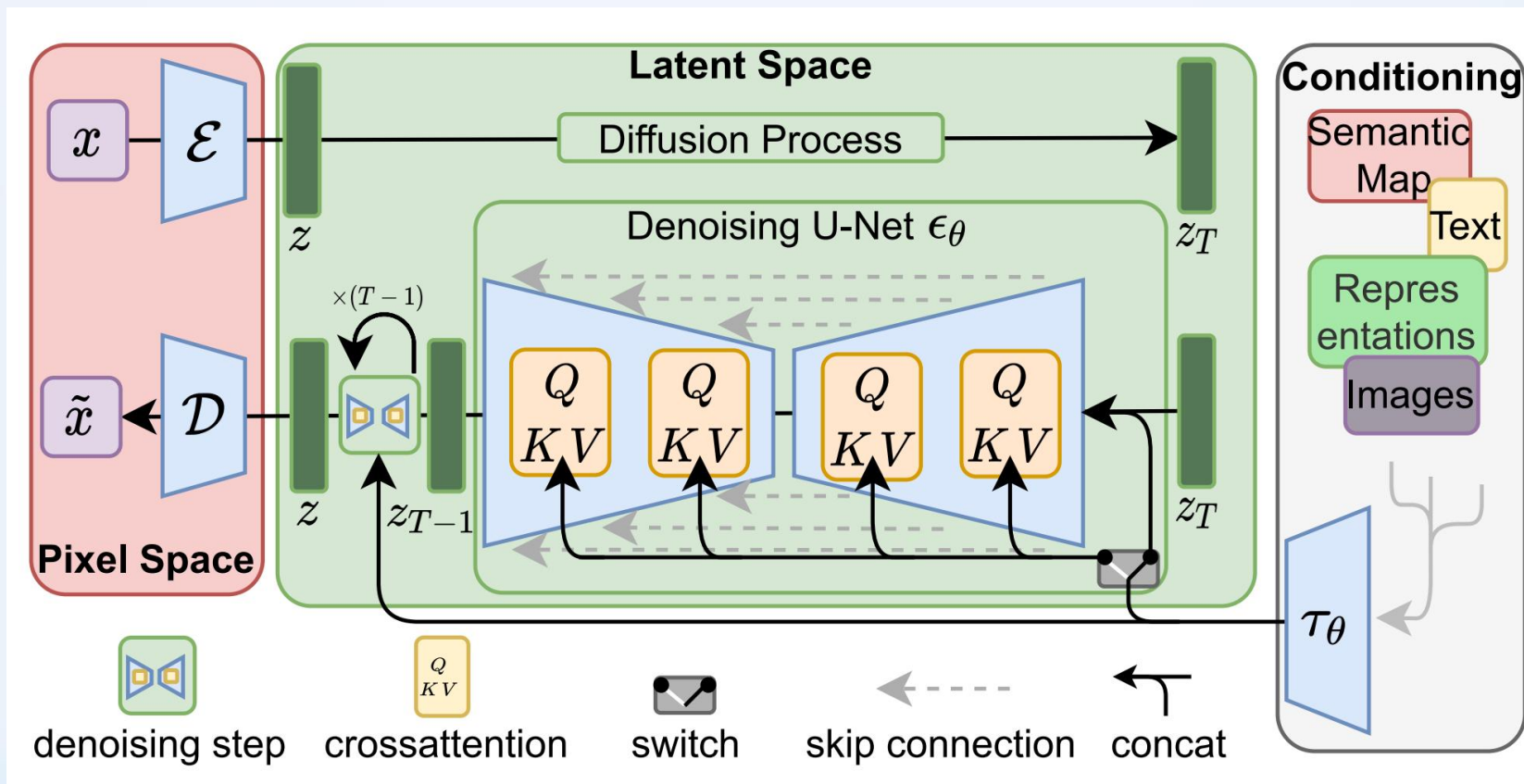
- 1: **repeat**
- 2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
- 3: $t \sim \text{Uniform}(\{1, \dots, T\})$
- 4: $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 5: Take gradient descent step on
 $\nabla_{\theta} \|\epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, t)\|^2$
- 6: **until** converged

Algorithm 2 Sampling

- 1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 2: **for** $t = T, \dots, 1$ **do**
- 3: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
- 4: $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$
- 5: **end for**
- 6: **return** $\mathbf{x}_0$

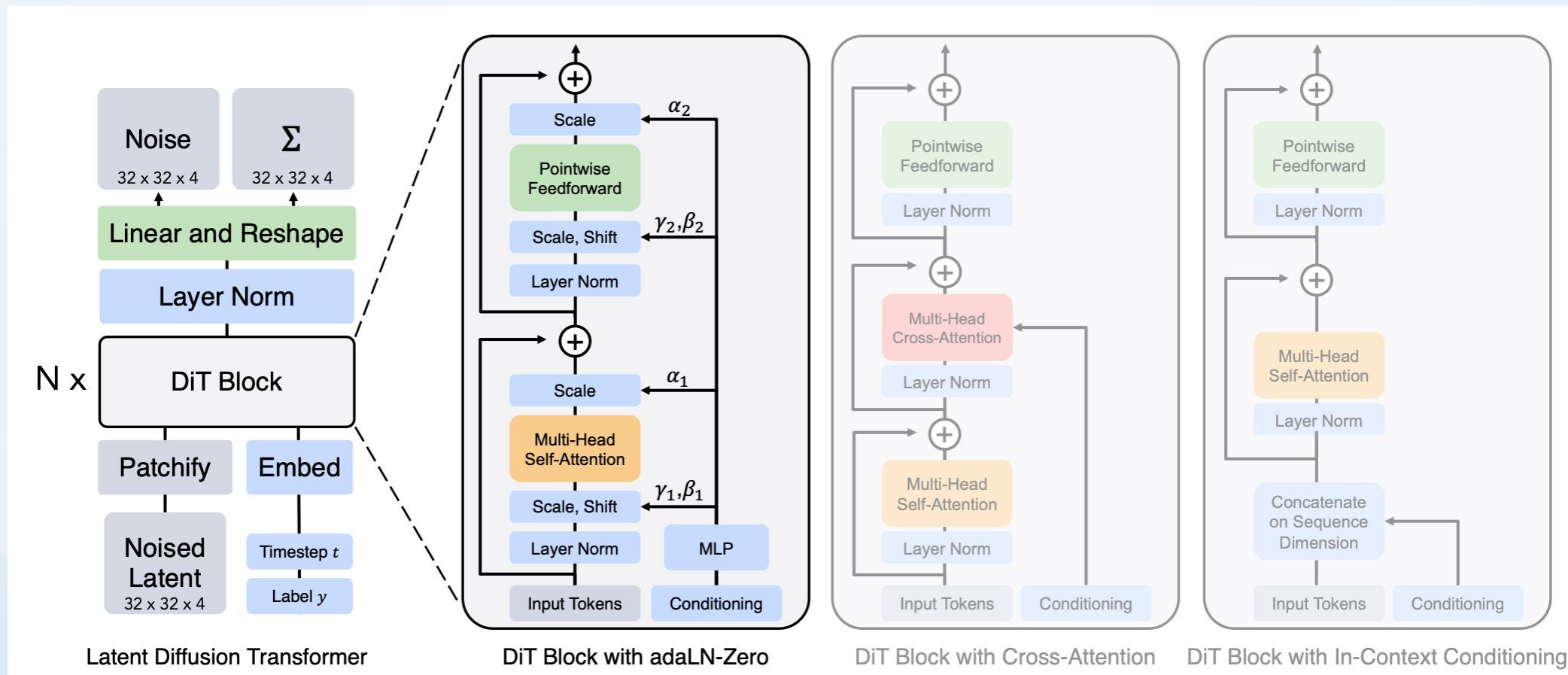
* Credit: [2020] Denoising Diffusion Probabilistic Models

Diffusion Models



* Credit: [2021] High-Resolution Image Synthesis with Latent Diffusion Models

Diffusion Models



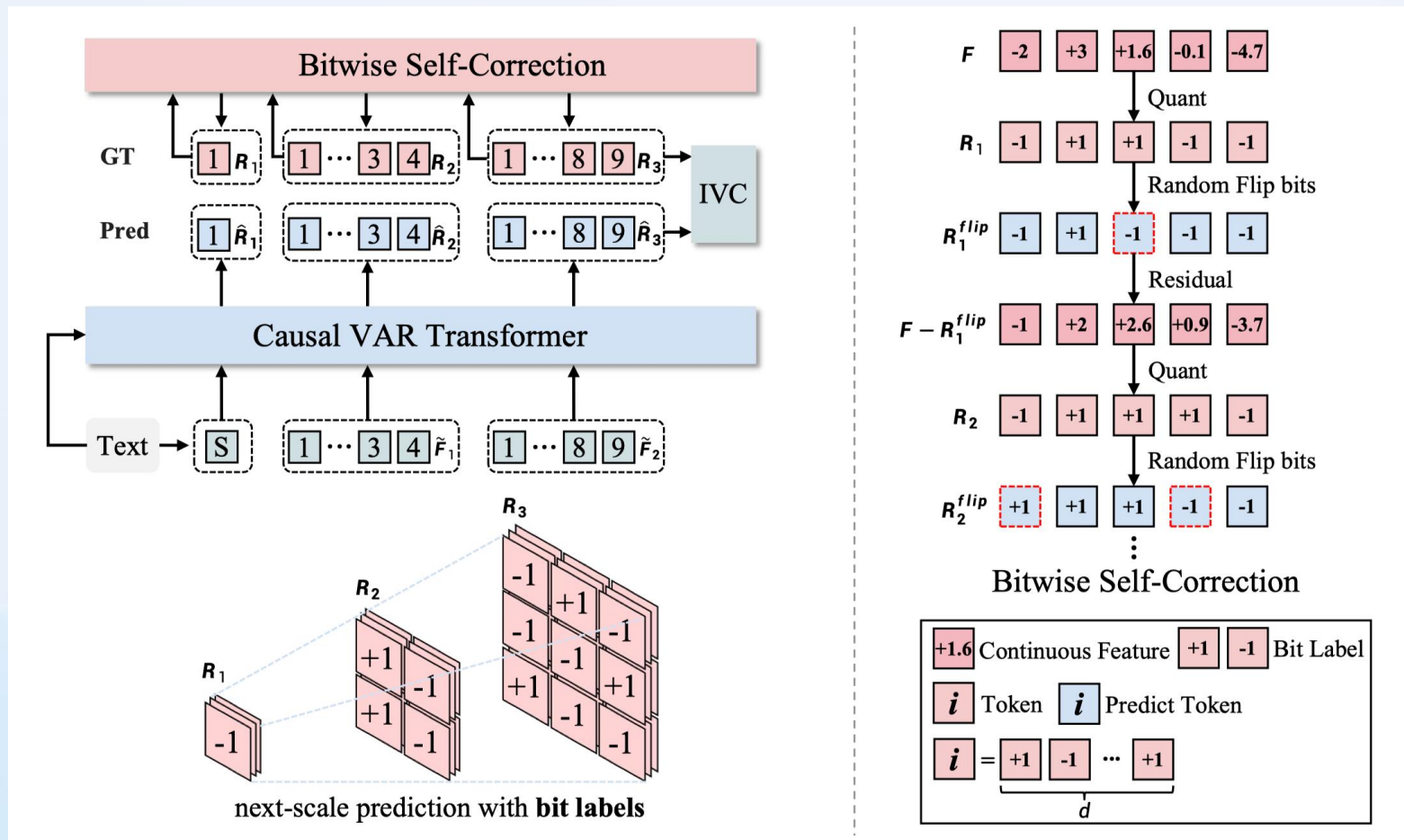
* Credit: [2022] Scalable Diffusion Models with Transformers

02

自回归文生图模型：Infinity

自回归文生图 Infinity

Infinity redefines VAR under a bitwise modeling framework

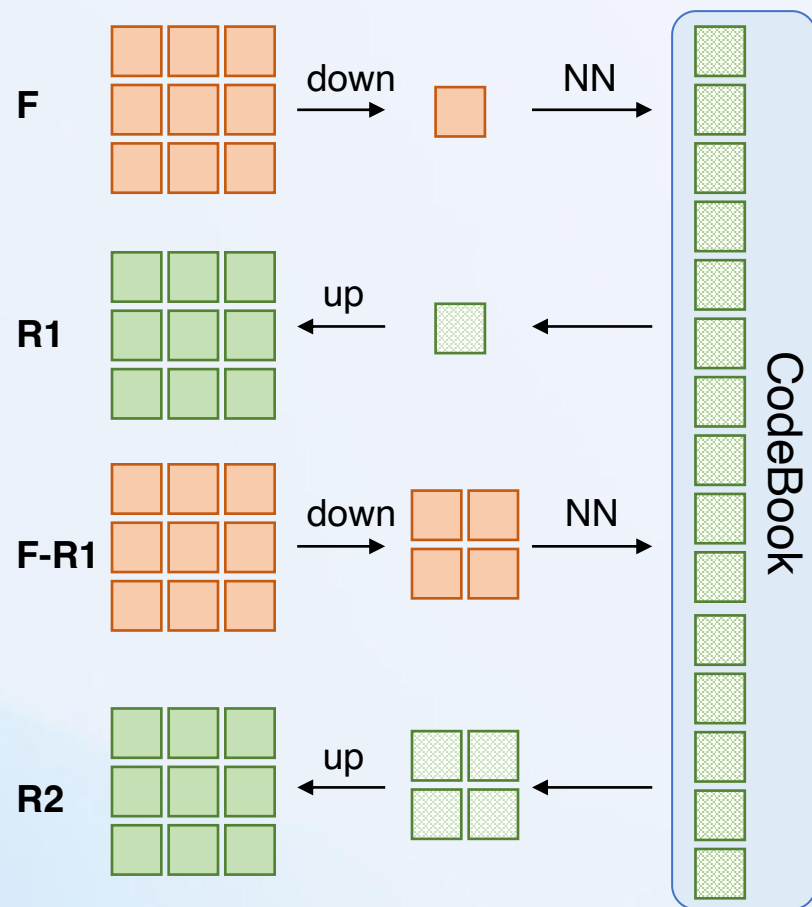


Bitwise modeling:

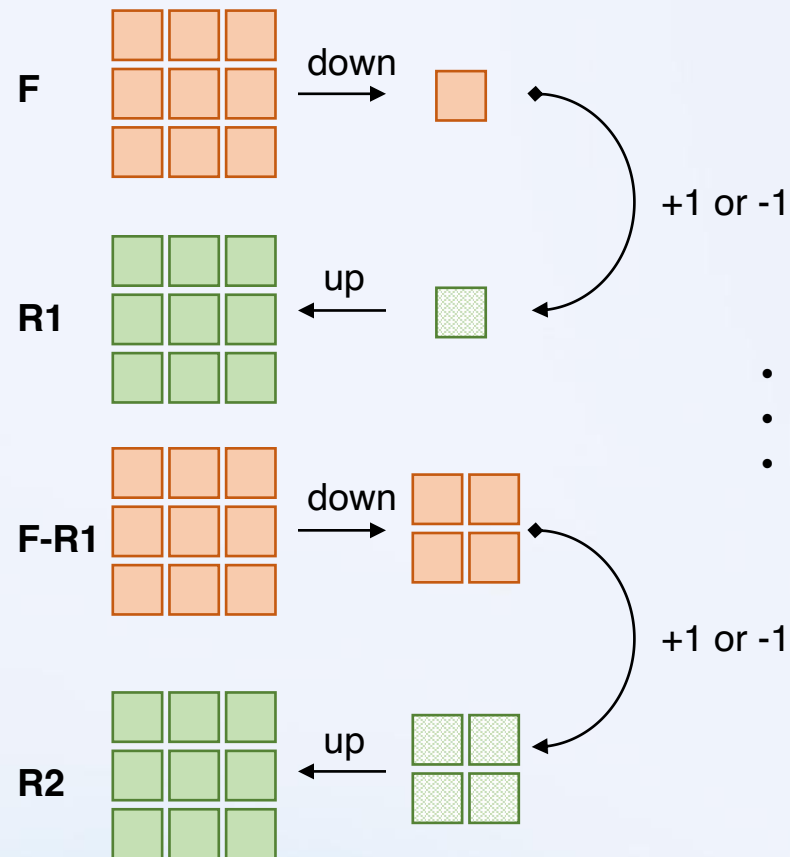
- Bitwise Tokenizer
- Infinite-Vocabulary Classifier
- Bitwise Self-Correction

Bitwise Tokenizer

An example for VQ and BSQ. We omit the channel dimension for simplicity



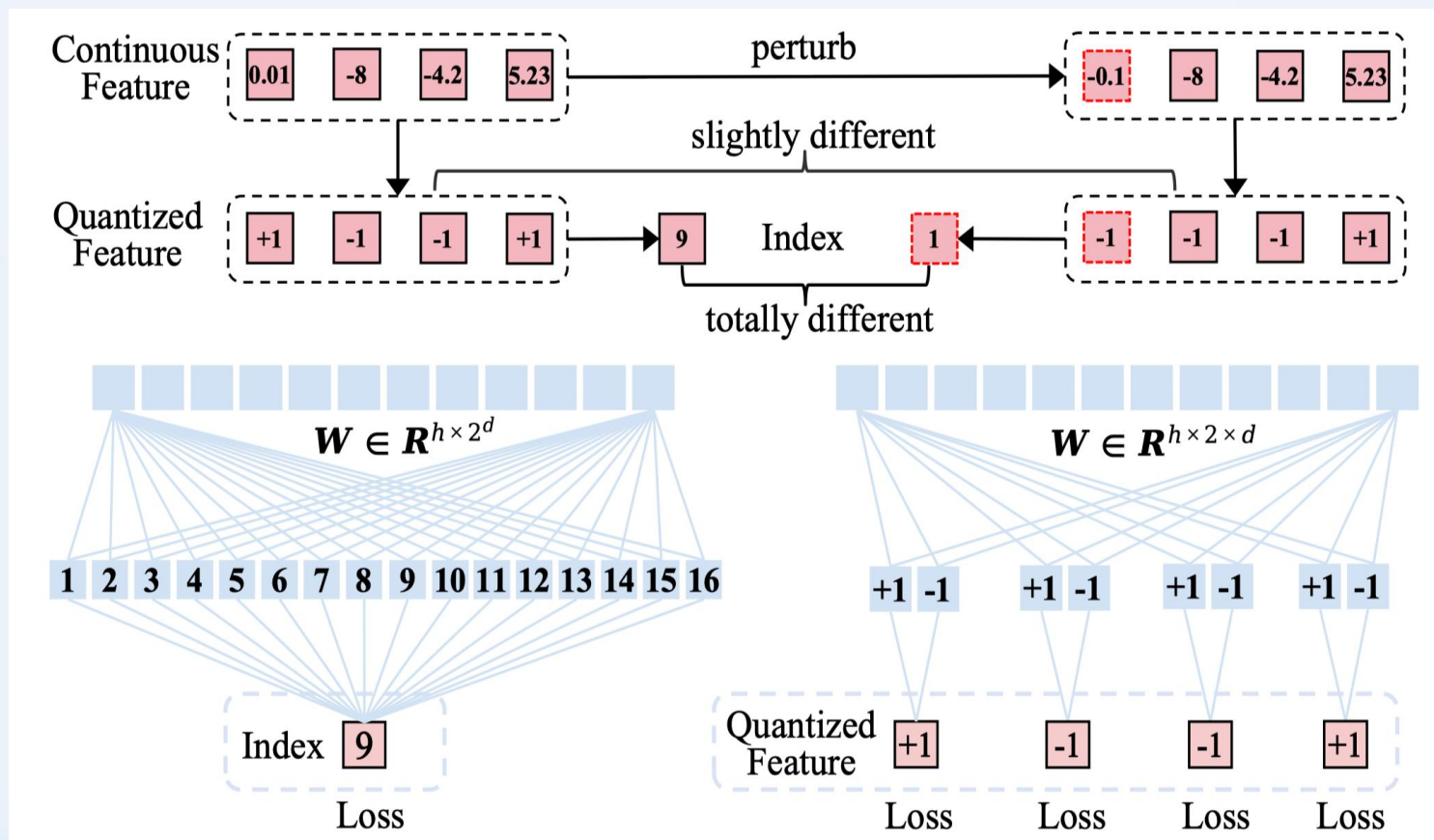
Vector Quantize



BSQ

- up is up sample
- down is down sample
- NN is nearest neighbour

Bitwise Tokenizer

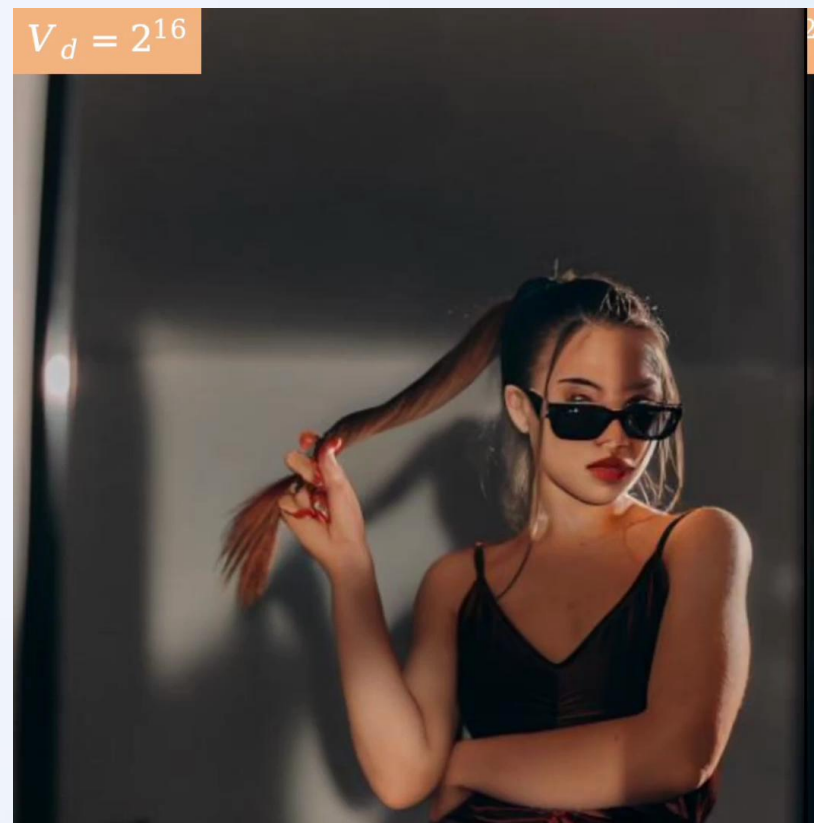


- Infinite-Vocabulary Classifier: predicts d bits instead of predicting 2^d indices
- Robust to slight perturbation and error

Bitwise Tokenizer

By Scaling Visual Tokenizer' s Vocabulary,
our tokenizer improves and even surpasses
continuous VAE of SD on ImageNet-rFID.

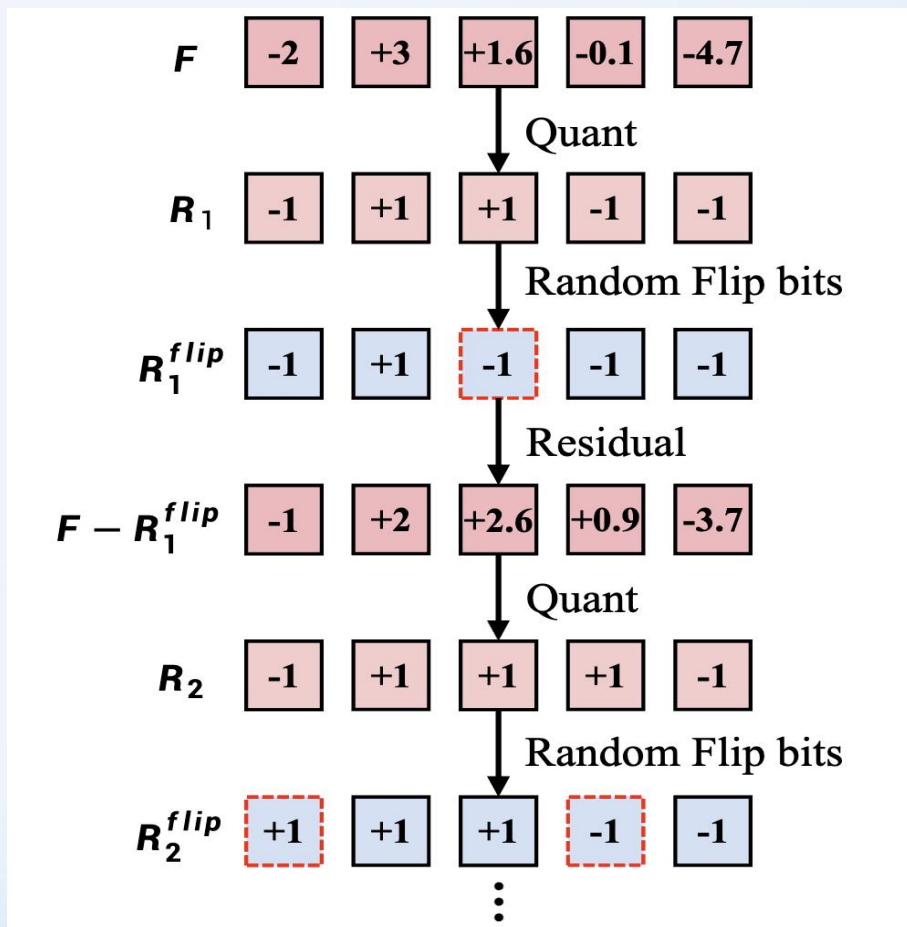
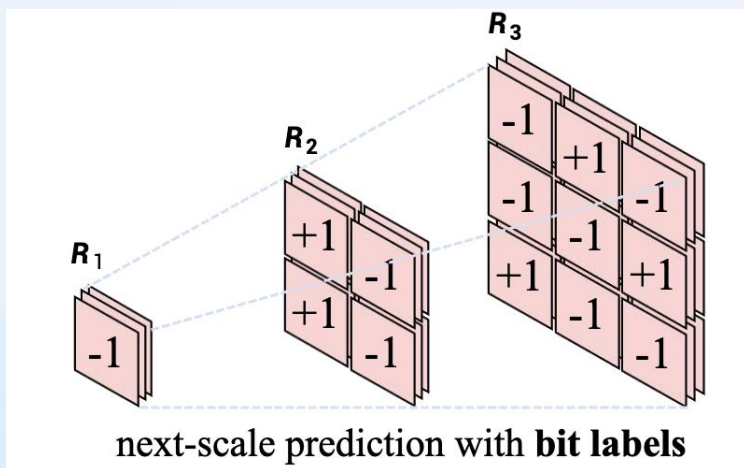
VAE (stride=16)	TYPE	IN-256 rFID↓	IN-512 rFID↓
$V_d = 2^{16}$	Discrete	1.22	0.31
$V_d = 2^{24}$	Discrete	0.75	0.30
$V_d = 2^{32}$	Discrete	0.61	0.23
$V_d = 2^{64}$	Discrete	0.33	0.15
SD VAE [49]	Contiguous	0.87	N/A



Bitwise Self-Correction

Bitwise Self-Correction to mitigate the issue of train-test discrepancy

- Random flip bits to imitates bitwise prediction errors
- Re-quantizing the residual features to auto-correct the previous errors



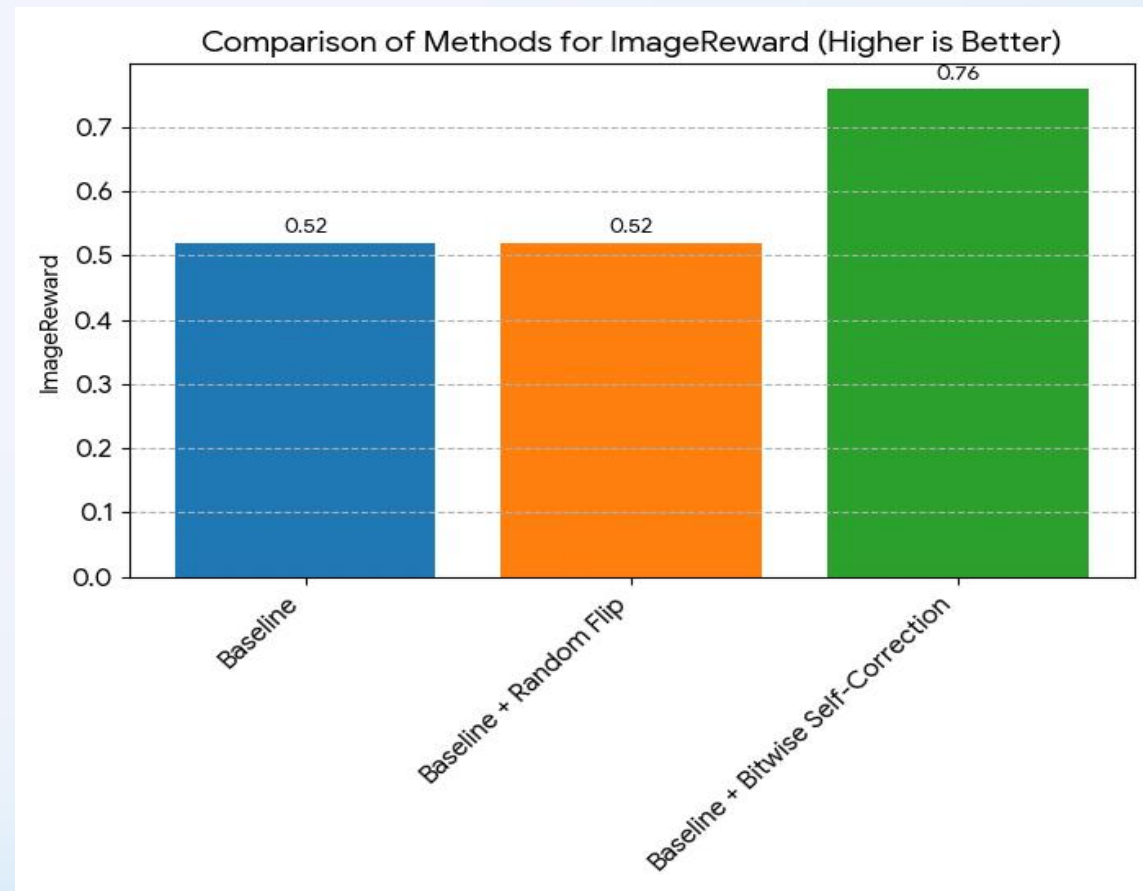
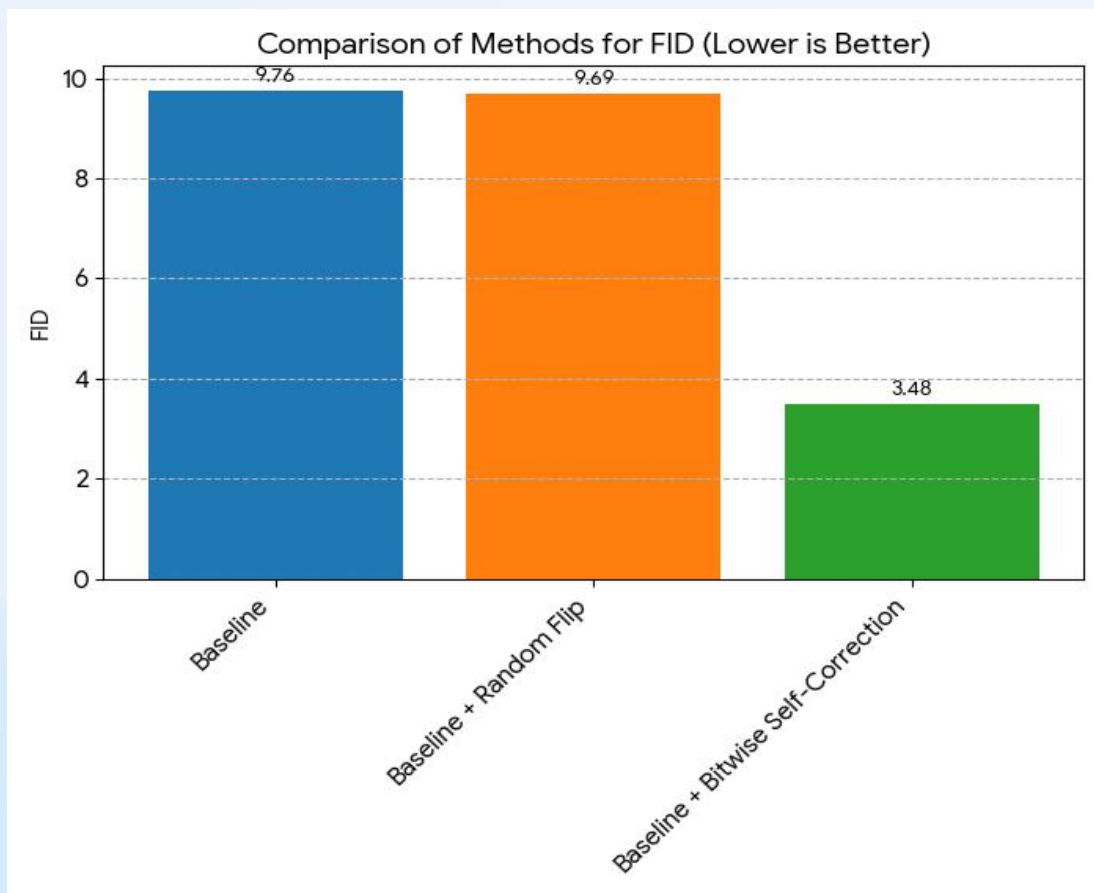
Bitwise Self-Correction

Using Bitwise Self-Correction generates better results (right)

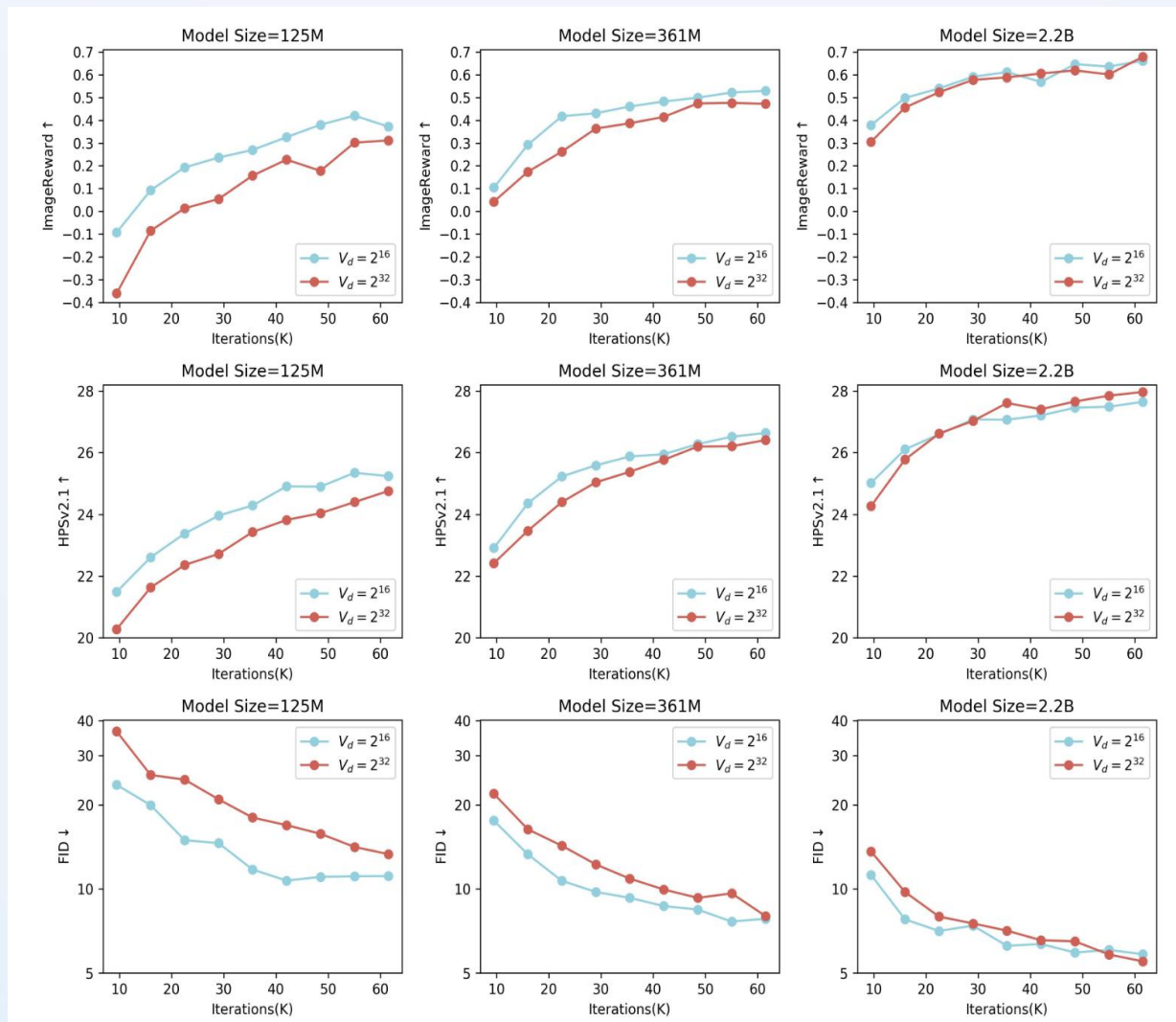


Bitwise Self-Correction

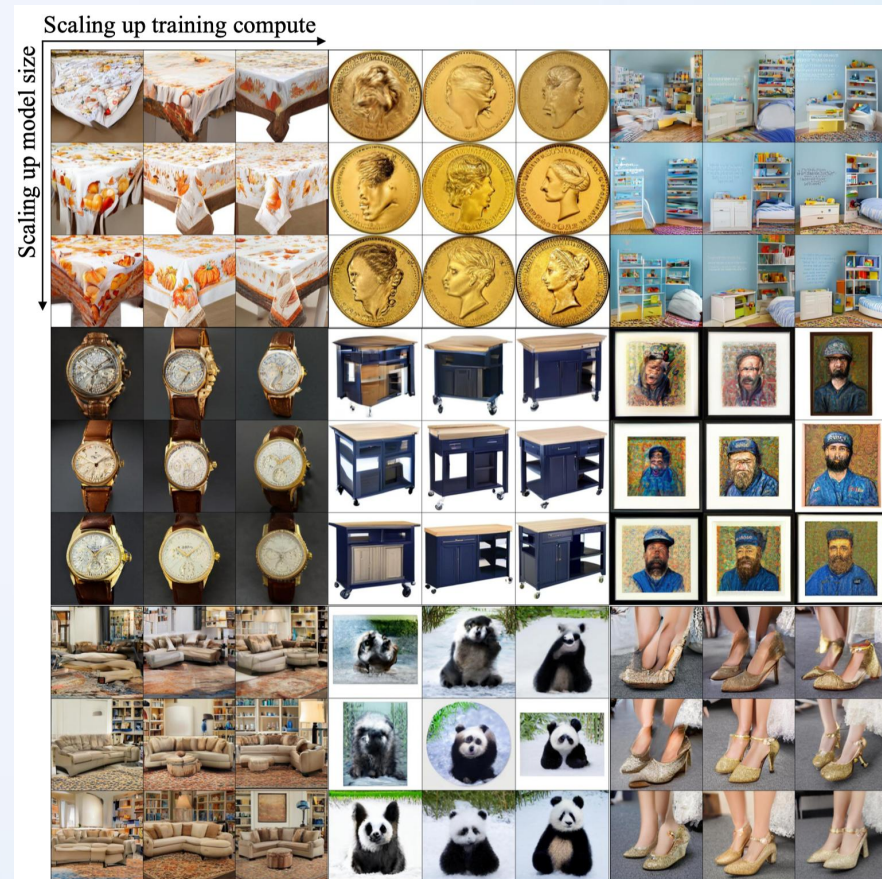
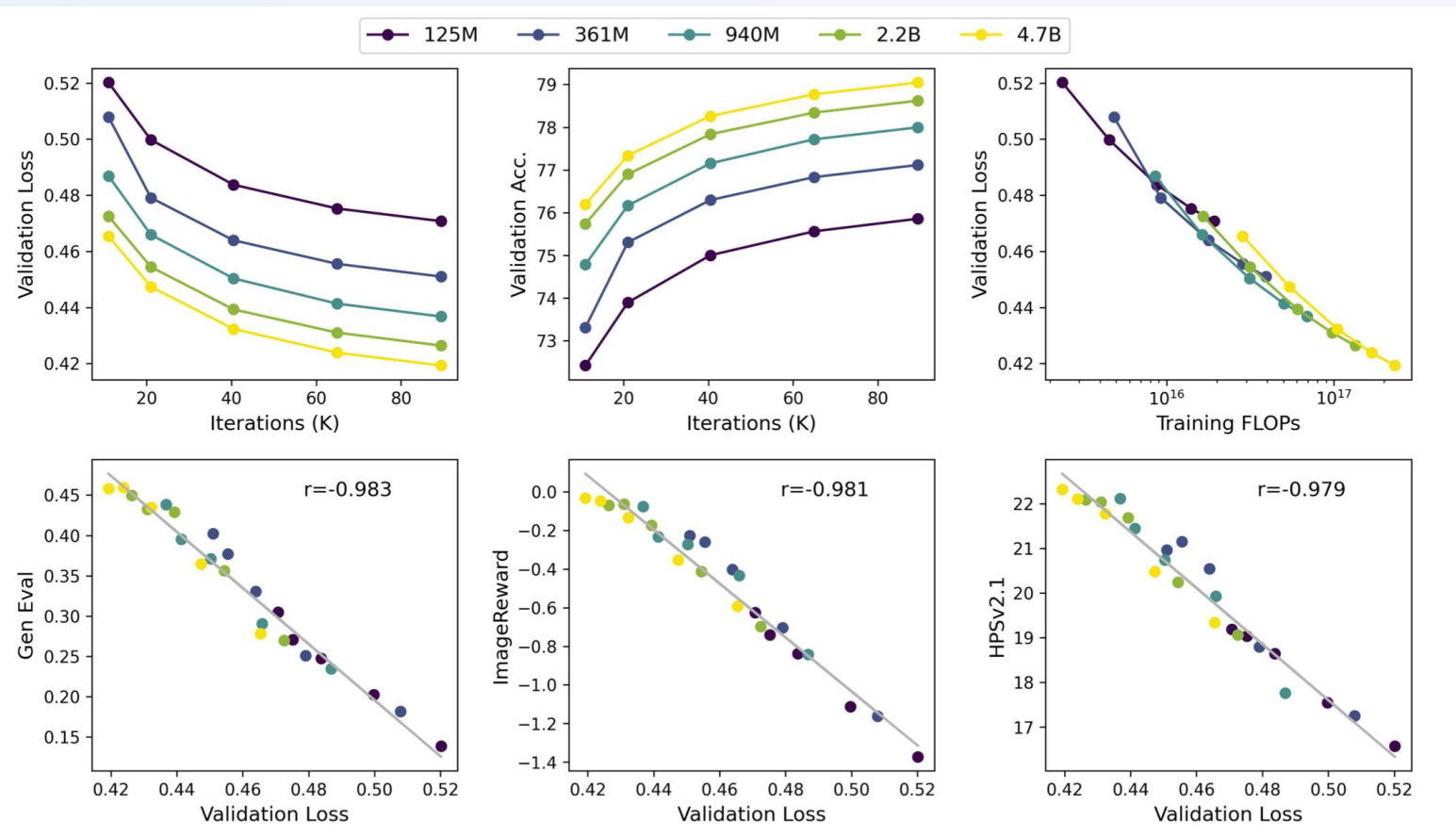
Qualitatively, Substantial advantages are observed after applying bitwise self-correction.



Scaling Vocabulary Benefits Generation



Scaling Model Benefits Generation

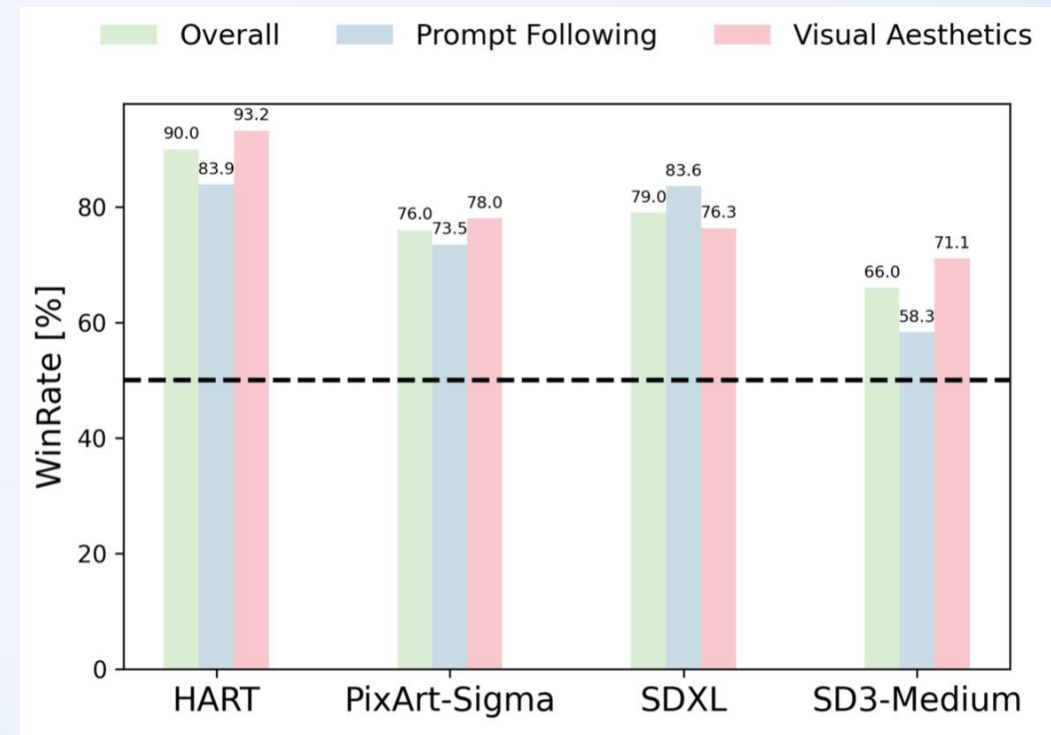




State-of-the-art Generation Results

Table 1: Evaluation on the GenEval [24] and DPG [29] benchmark. † result is with prompt rewriting.

Methods	# Params	GenEval↑				DPG↑		
		Two Obj.	Position	Color Attri.	Overall	Global	Relation	Overall
Diffusion Models								
LDM [49]	1.4B	0.29	0.02	0.05	0.37	-	-	-
SDv1.5 [49]	0.9B	0.38	0.04	0.06	0.43	74.63	73.49	63.18
PixArt-alpha [13]	0.6B	0.50	0.08	0.07	0.48	74.97	82.57	71.11
SDv2.1 [49]	0.9B	0.51	0.07	0.17	0.50	77.67	80.72	68.09
DALL-E 2 [45]	6.5B	0.66	0.10	0.19	0.52	-	-	-
DALL-E 3 [7]	-	-	-	-	0.67 [†]	90.97	90.58	83.50
SDXL [43]	2.6B	0.74	0.15	0.23	0.55	83.27	86.76	74.65
PixArt-Sigma [12]	0.6B	0.62	0.14	0.27	0.55	86.89	86.59	80.54
SD3 (d=24) [21]	2B	0.74	0.34	0.36	0.62	-	-	84.08
SD3 (d=38) [21]	8B	0.89	0.34	0.47	0.71	-	-	-
AutoRegressive Models								
LlamaGen [55]	0.8B	0.34	0.07	0.04	0.32	-	-	65.16
Chameleon [59]	7B	-	-	-	0.39	-	-	-
HART [58]	732M	-	-	-	0.56	-	-	80.89
Show-o [70]	1.3B	0.80	0.31	0.50	0.68	-	-	67.48
Emu3 [66]	8.5B	0.81 [†]	0.49 [†]	0.45 [†]	0.66 [†]	-	-	81.60
Infinity	2B	0.85 [†]	0.49 [†]	0.57 [†]	0.73 [†]	93.11	90.76	83.46



Text-to-Image Arena

Black Forest Labs	FLUX.1 [pro]	1098	-5/+4	42,725	Aug 2024
Ideogram	Ideogram v2	1076	-4/+4	41,223	Aug 2024
Midjourney	Midjourney v6.1	1076	-4/+4	42,611	Jul 2024
Midjourney	Midjourney v7 Alpha	1075	-7/+7	13,115	Apr 2025
Black Forest Labs	FLUX.1 [dev]	1074	-4/+4	41,827	Aug 2024
Ideogram	Ideogram v1	1065	-4/+4	41,568	Feb 2024
Midjourney	Midjourney v6	1060	-4/+4	52,416	Dec 2023
HiDream	HiDream-I1-Dev	1059	-6/+7	12,901	Apr 2025
Adobe	Firefly Image 4	1053	-10/+8	7,317	Apr 2025
Bytedance	Infinity 8B	1051	-6/+6	14,073	Feb 2025
Ideogram	Ideogram v2 Turbo	1048	-5/+4	41,892	Aug 2024
Stability.ai	Stable Diffusion 3.5 Large	1036	-4/+4	41,158	Oct 2024
Luma Labs	Luma Photon	1036	-6/+6	15,851	Dec 2024
Stability.ai	Stable Diffusion 3.5 Large Turbo	1036	-4/+4	40,699	Oct 2024
Leonardo.Ai	Phoenix 1.0 Ultra	1035	-7/+7	14,299	Dec 2024
Stability.ai	Stable Diffusion 3 Large	1024	-4/+4	50,052	Feb 2024

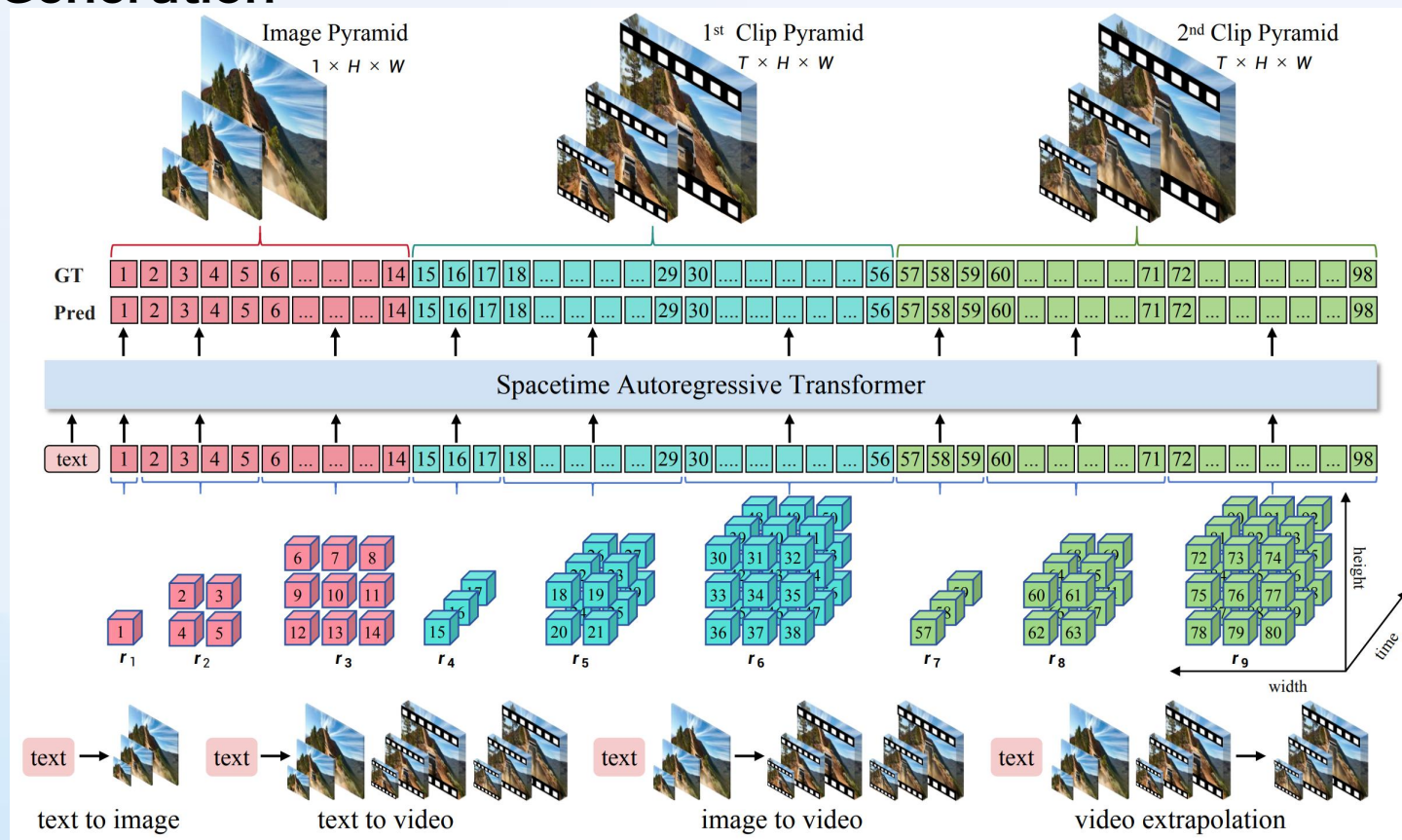
Our open-source Infinity-8B is comparable to open or close-source leading diffusion models

03

自回归视频生成模型：InfinityStar

Extend Infinity to T2V: InfinityStar

Unified Spacetime AutoRegressive Modeling for Visual Generation



InfinityStar:

- Novel spacetime pyramid modeling that unifies diverse visual generation tasks: T2I, T2V, I2V, V2V
- Outperform existing autoregressive text-to-video models and matching leading diffusion models
- Over 10x speedup regarding the inference latency

"InfinityStar: Unified SpaceTime AutoRegressive Modeling for Visual Generation" NeurIPS 2025 Oral

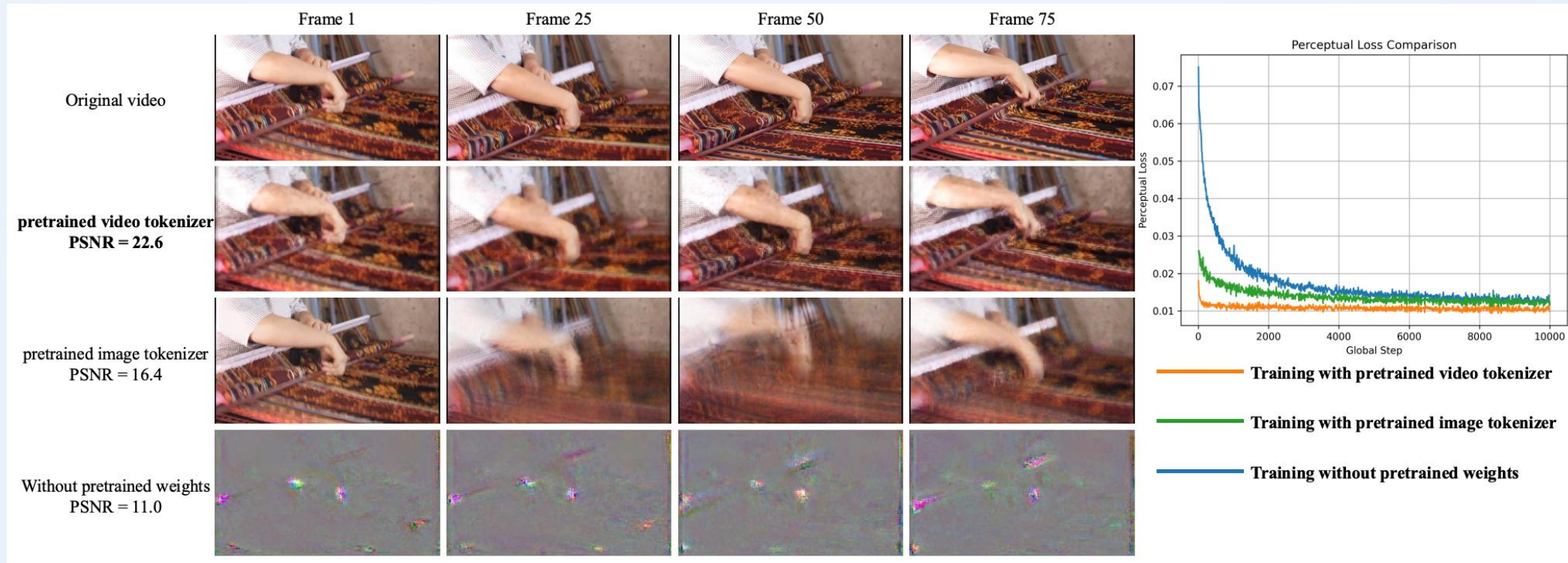
Challenges from T2I to T2V

Poor reconstruction of video discrete tokenizer

- There's no multi-scale discrete visual tokenizers
- Train a video tokenizer from scratch is extremely time-consuming and converges slow
- Imbalanced information distribution among scales (Most info is concentrated in the last few scales)

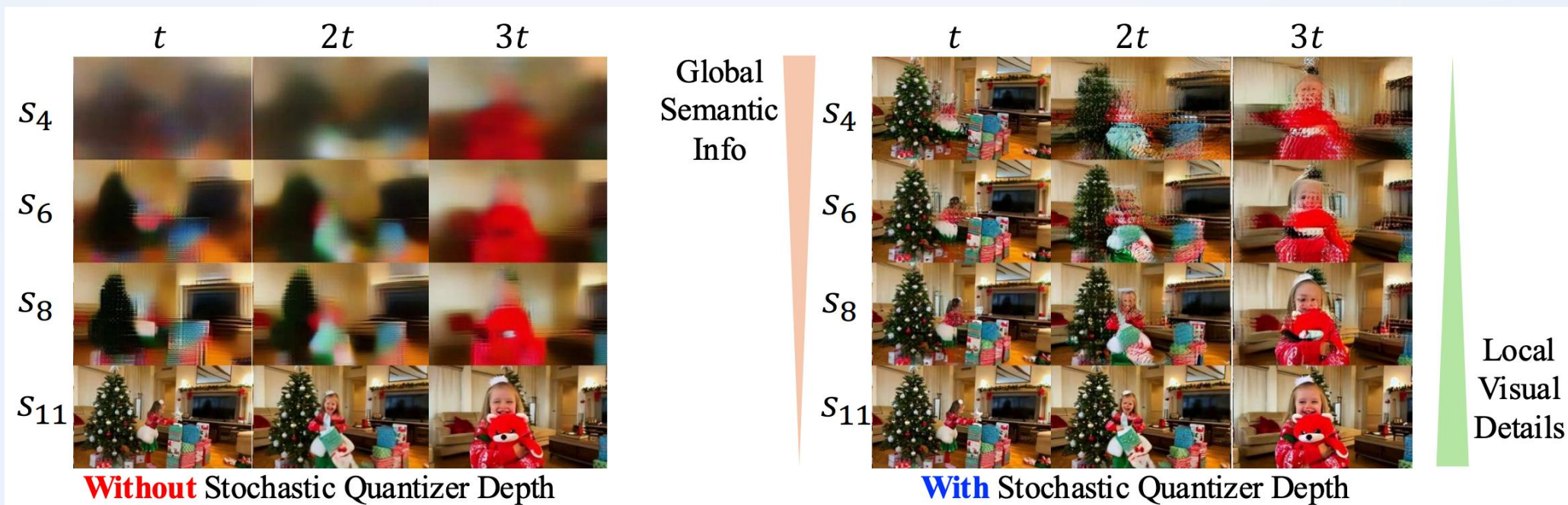


InfinityStar: Video Tokenizer



Training with pretrained video tokenizer converges significantly faster than that of training from scratch or loading image tokenizer

InfinityStar: Video Tokenizer



- In the multi-scale visual tokenizer, earlier scales correspond to global semantics, later ones correspond to local details.
- **Stochastic Quantizer Depth** significantly improves the reconstruction quality of early scales.

Challenges from T2I to T2V

How to incorporate the temporal dimension?

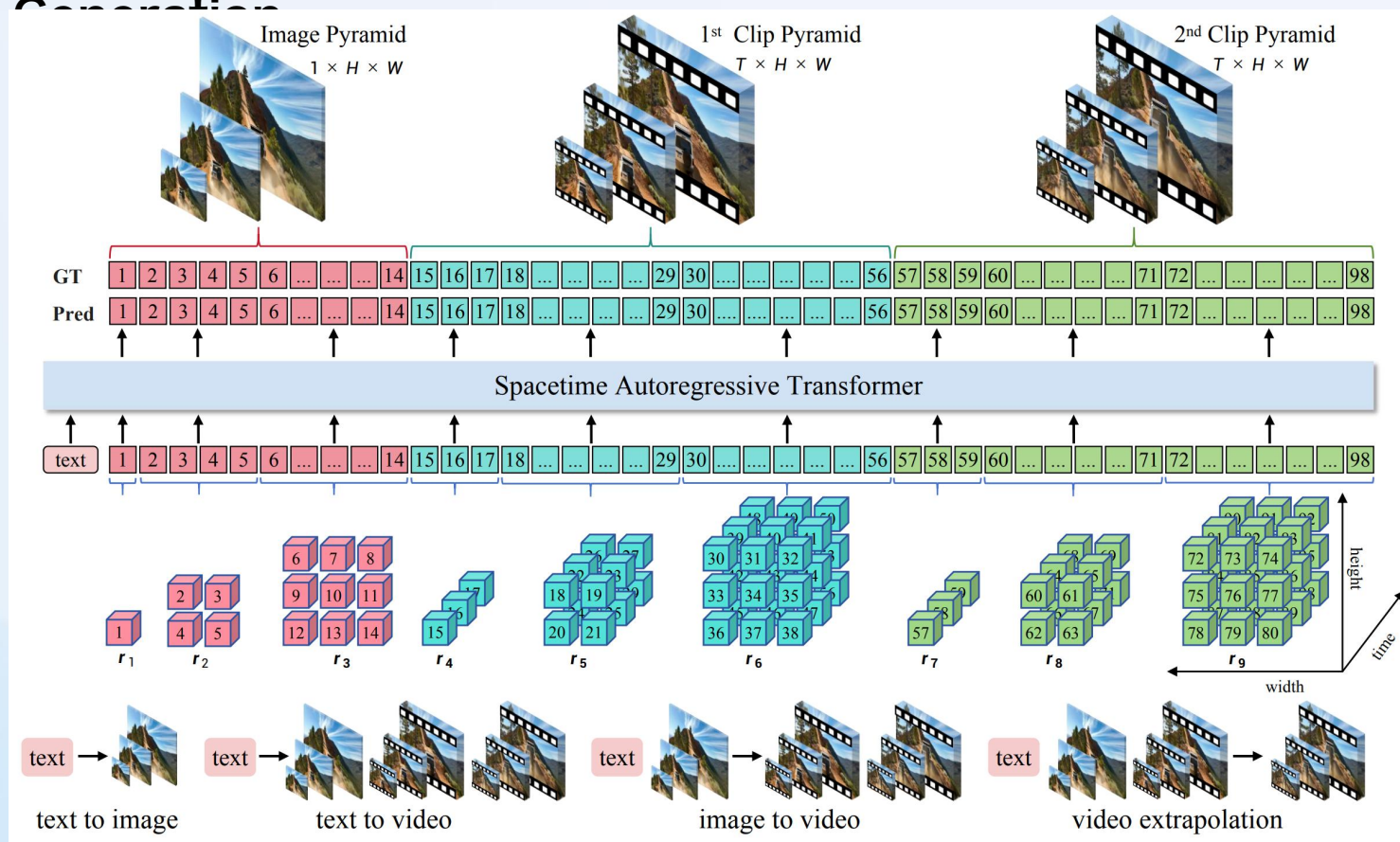
Method	Result	Work
Grow uniformly from (1, 1, 1) to (T, H, W)	flickering	
Pseudo-Spacetime Pyramid: Grow from (T, 1, 1) to (T, H, W)	lack details	
Spacetime Pyramid	good	



The video shows a cable car system with a tower in the foreground, situated on a mountainous area with lush greenery. In the background, there is a cityscape with buildings and a river, partially obscured by fog. A cable car moves from the right towards the left.

Spacetime Autoregressive Transformer

Overview: Spacetime Pyramid Modeling for Unified Generation



Each video is decomposed into sequential clips
 c_1 is the first frame, c_n is a video clip

$$\{c_1, c_2, \dots, c_N\}$$

For inner clip predictions, all scales in the pyramid

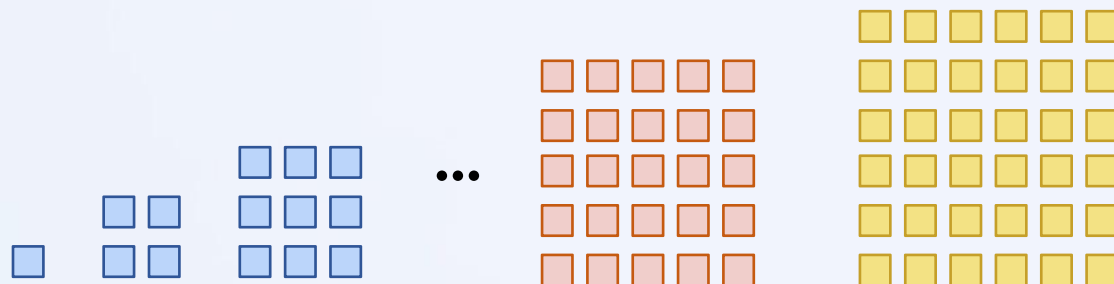
$$p(r_1^1, \dots, r_K^1) = \prod_{k=1}^K p(r_k^1 | r_1^1, \dots, r_{k-1}^1, \psi(t))$$

For inter-clip predictions, clips are generated autoregressively conditioned on prior clip predictions

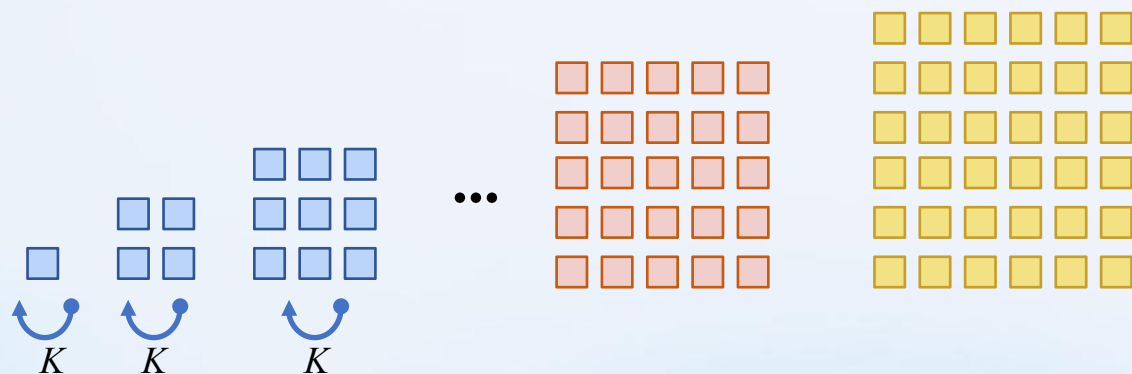
$$p(r_1^1, \dots, r_K^N) = \prod_{c=1}^N \prod_{k=1}^K p(r_k^c | r_1^1, \dots, r_{k-1}^c, \psi(t))$$

Spacetime Autoregressive Transformer

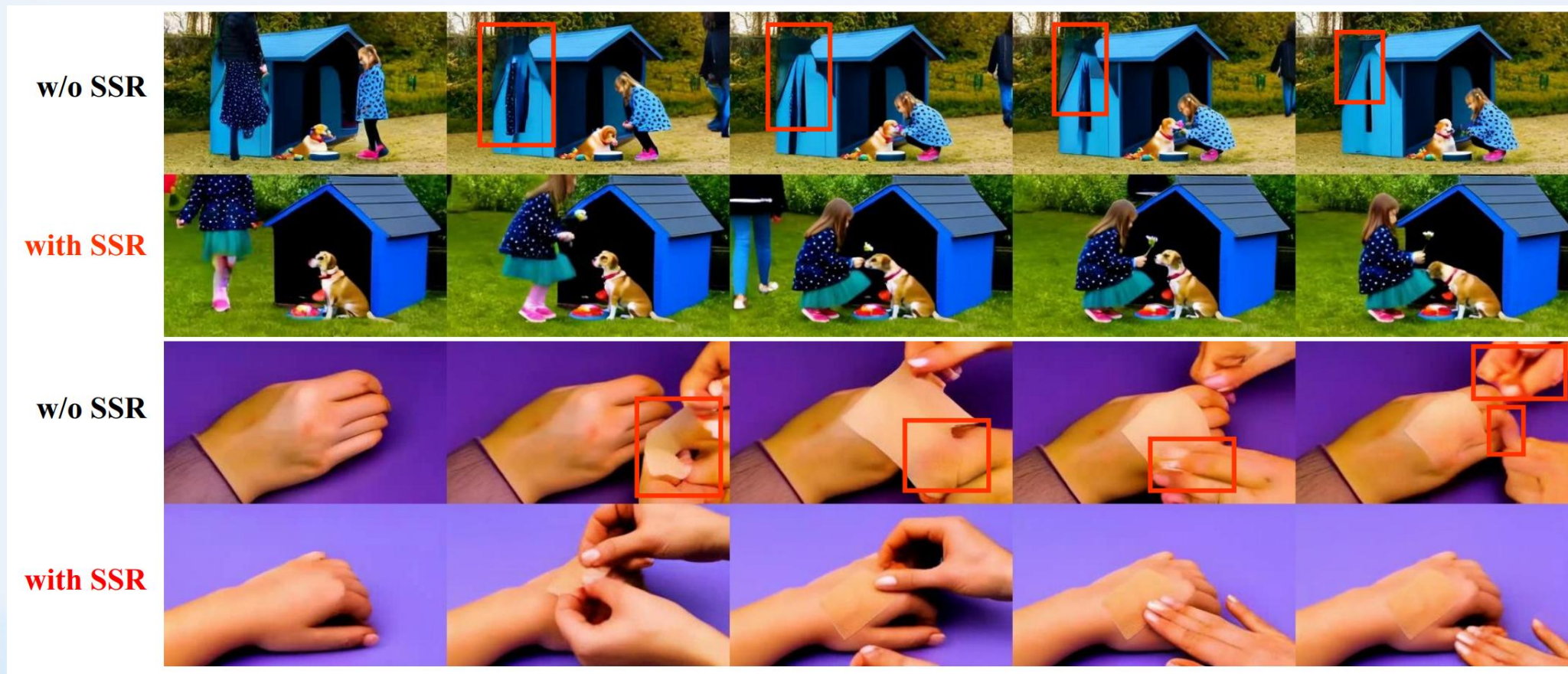
Vanilla Next-Scale Prediction



Semantic Scale Repetition

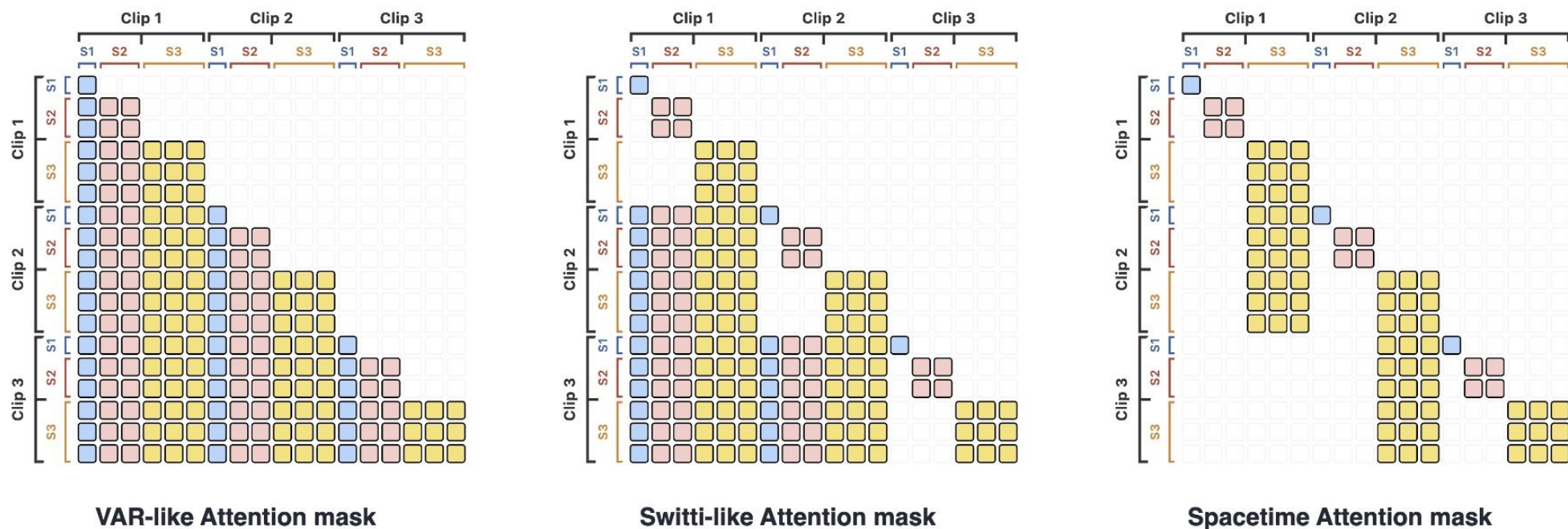


Spacetime Autoregressive Transformer



- Semantic Scale Repetition (SSR) greatly improves structure stability and motion quality

Spacetime Autoregressive Transformer: **Spacetime Sparse Attention**

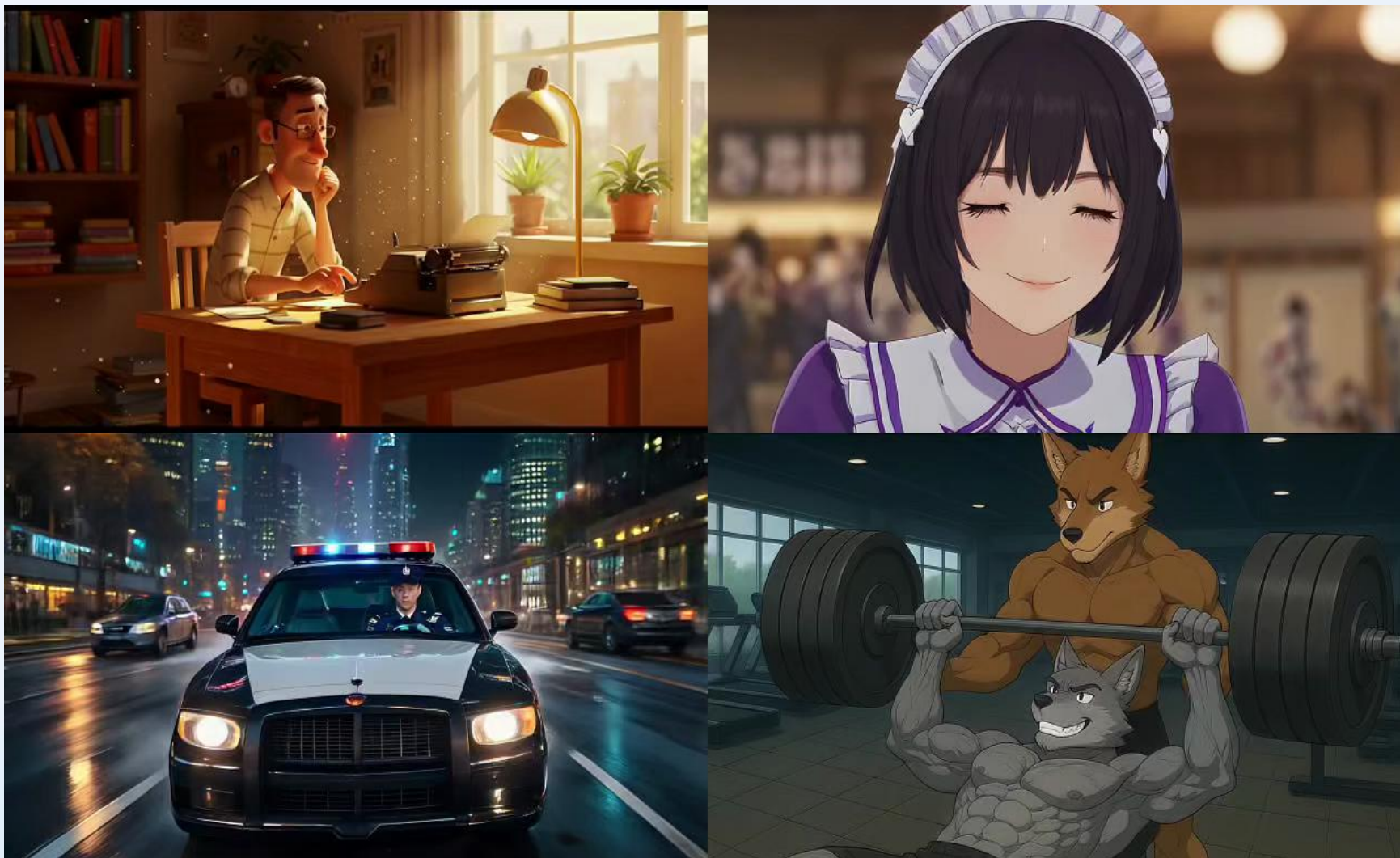


- Autoregressive video generation struggles with the high computational costs of long context
- Spacetime Sparse Attention drastically reduces computational overhead while delivering better performance

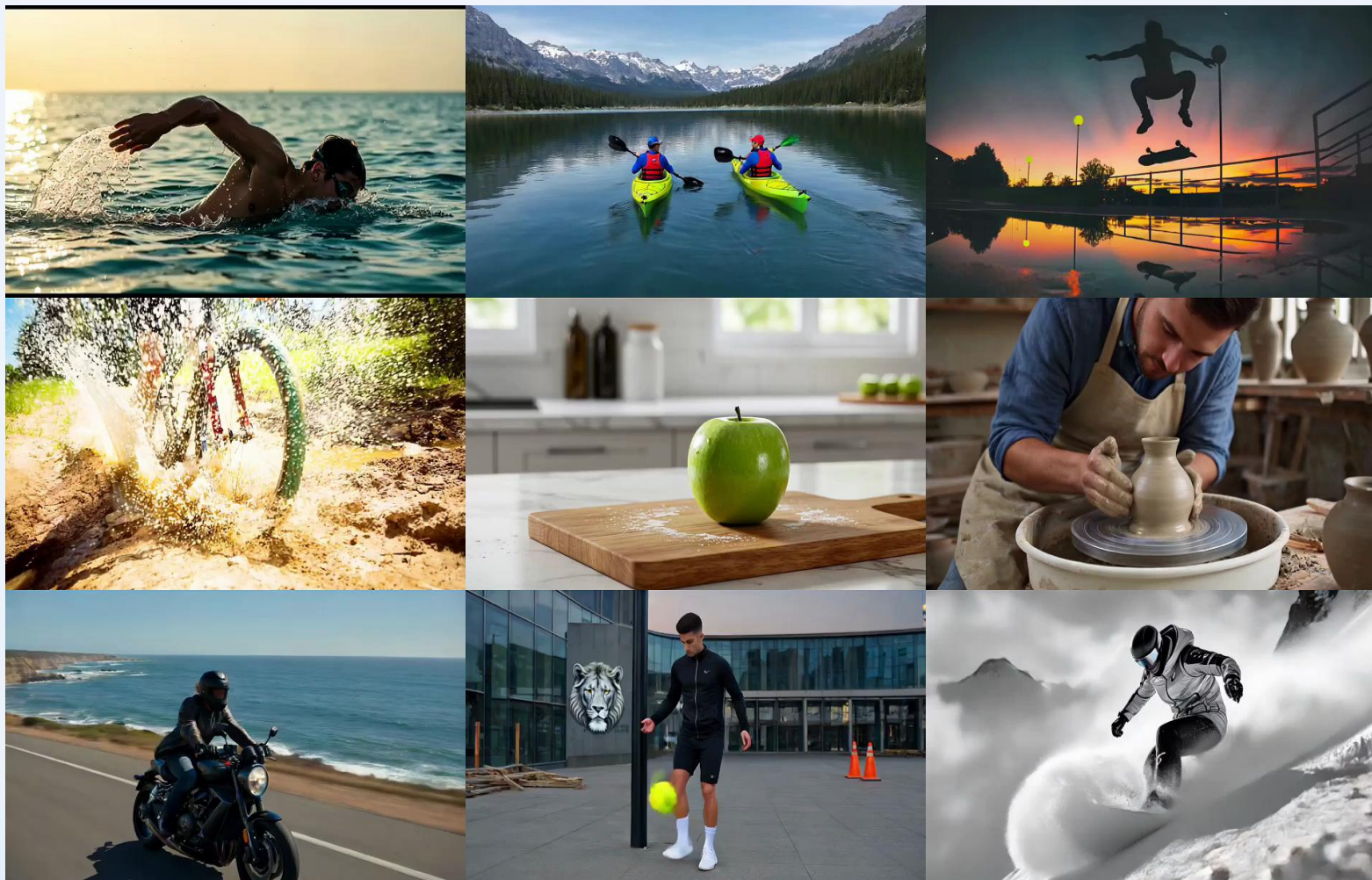
Qualitative Results



Qualitative Results



Qualitative Results



Quantitative Results: T2I benchmark Evaluation

Table 1: Evaluation on the GenEval [14] and DPG [16] benchmark. † result is with prompt rewriting or self-CoT.

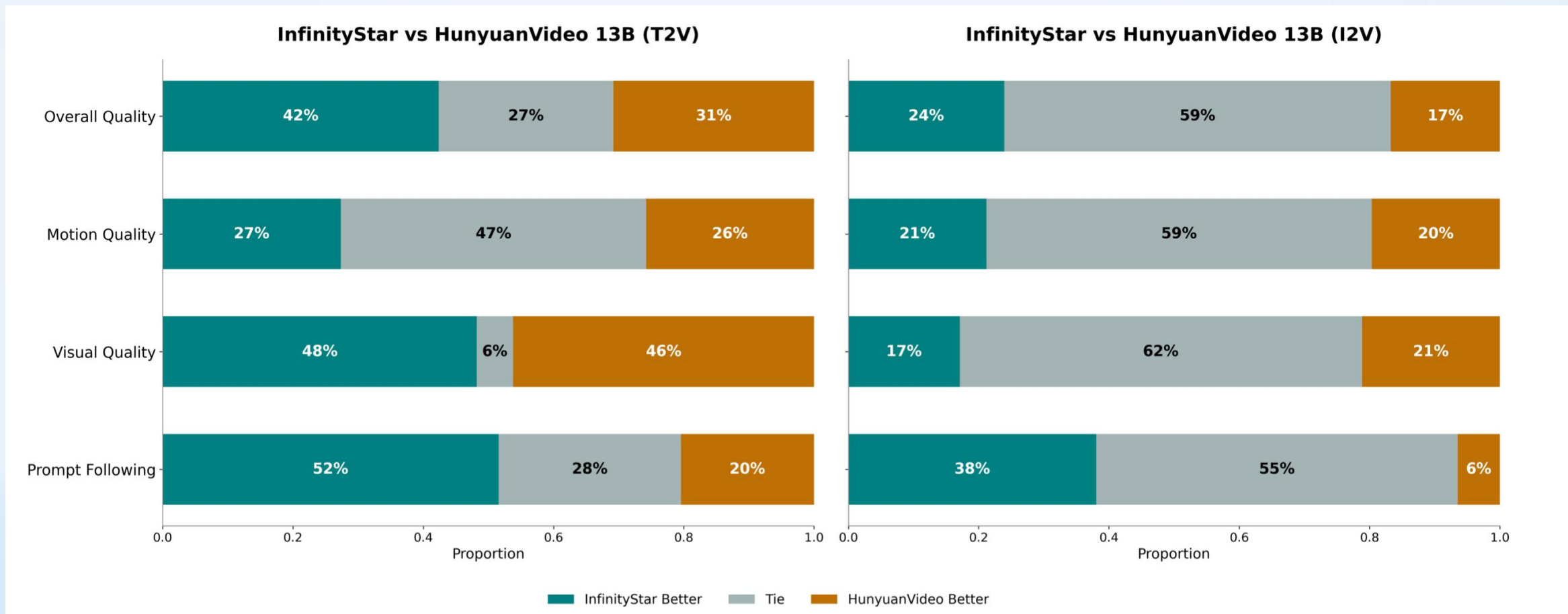
Methods	# Params	GenEval↑				DPG↑		
		Two Obj.	Position	Color Attri.	Overall	Global	Relation	Overall
Diffusion Models								
SDXL [24]	2.6B	0.74	0.15	0.23	0.55	83.27	86.76	74.7
PixArt-Sigma [5]	0.6B	0.62	0.14	0.27	0.55	86.89	86.59	80.5
SD3 (d=38) [11]	8B	0.89	0.34	0.47	0.71	-	-	-
Goku [6]	2B	-	-	-	0.76 [†]	-	-	83.6
Transfusion [51]	7.3B	-	-	-	0.63	-	-	-
SANA-1.0 [37]	1.6B	-	-	-	0.66	-	-	84.8
FLUX-dev [21]	12B	-	-	-	0.67	-	-	84.0
FLUX-schnell [21]	12B	-	-	-	0.71	-	-	84.8
AutoRegressive Models								
LlamaGen [26]	0.8B	0.34	0.07	0.04	0.32			65.2
Chameleon [27]	7B	-	-	-	0.39	-	-	-
Show-o [38]	1.3B	0.80	0.31	0.50	0.68	-	-	67.5
Liquid [36]	7B	0.73	0.17	0.37	0.55	-	-	-
UniTok [22]	7B	0.71	0.26	0.45	0.59	-	-	-
Janus [35]	1.3B	0.68	0.46	0.42	0.61	-	-	-
Emu3 [34]	8B	0.81 [†]	0.49 [†]	0.45 [†]	0.66 [†]	-	-	81.6
Fluid [12]	10.5B	0.83	0.39	0.51	0.69	-	-	-
NextStep-1 [28]	14B	-	-	-	0.73 [†]	-	-	85.28
Infinity [15]	2B	0.85 [†]	0.49 [†]	0.57 [†]	0.73 [†]	93.11	90.76	83.46
InfinityStar-T2I	8B	0.90 [†]	0.62 [†]	0.67 [†]	0.79 [†]	91.68	91.87	86.55

Quantitative Results: T2V benchmark Evaluation

Table 2: Evaluation on the VBench benchmark. † result is with prompt rewriting.

Models	# Params	Human Action	Scene	Multiple Objects	Appear. Style	Quality Score	Semantic Score	Overall
Diffusion Models								
AnimateDiff-V2	1.5B	92.60	50.19	36.88	22.42	82.90	69.75	80.27
VideoCrafter-2.0[4]	1.5B	95.00	<u>55.29</u>	40.66	25.13	82.20	73.42	80.44
OpenSora V1.2[50]	1.1B	85.80	42.47	58.41	23.89	80.71	73.30	79.23
Show-1[44]	6B	95.60	47.03	45.47	23.06	80.42	72.98	78.93
Gen-3 [13]	-	96.40	54.57	53.64	24.31	84.11	75.17	82.32
CogVideoX-5B[40]	5B	99.40	53.20	62.11	<u>24.91</u>	82.75	77.04	81.61
HunyuanVideo[19]	13B	94.40	53.88	68.55	19.80	85.09	75.82	83.24
Goku[6]	2B	97.60	57.08	<u>79.48</u>	23.08	<u>85.60</u>	81.87	84.85
Wan 2.1[32]	14B	<u>98.80</u>	53.67	81.44	21.13	85.64	<u>80.95</u>	<u>84.70</u>
AutoRegressive Models								
Nova[10]†	0.6B	95.20	54.06	77.52	20.92	80.39	79.05	80.12
Emu3[34]	8B	77.71	37.11	44.64	20.92	84.09	68.43	80.96
InfinityStar†	8B	96.43	52.08	<u>78.66</u>	21.81	<u>84.73</u>	<u>79.78</u>	<u>83.74</u>

Human Evaluation

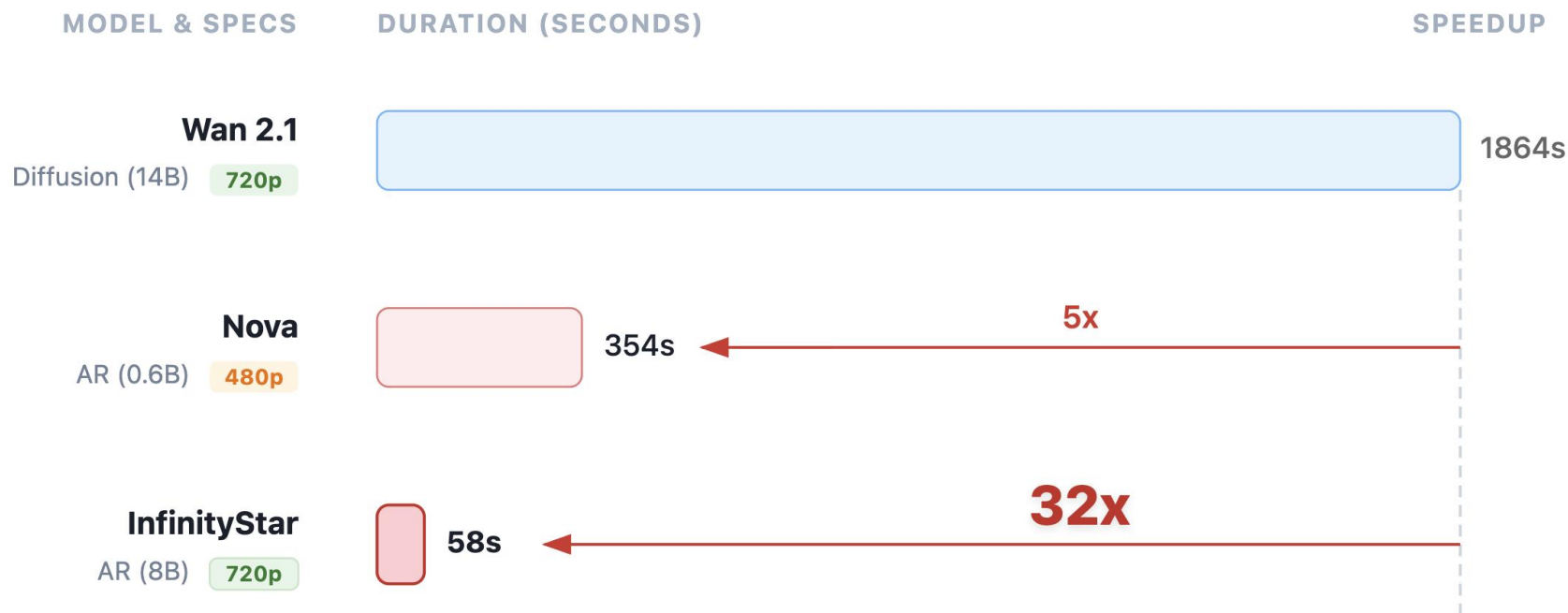


Human evaluation comparing InfinityStar (8B) with Hunyuan Video 1.0 (13B)

Generation Latency

Computational Efficiency Comparison

Generation Latency (Lower is Better) on 81 Frames



Interactive Long Video Generation

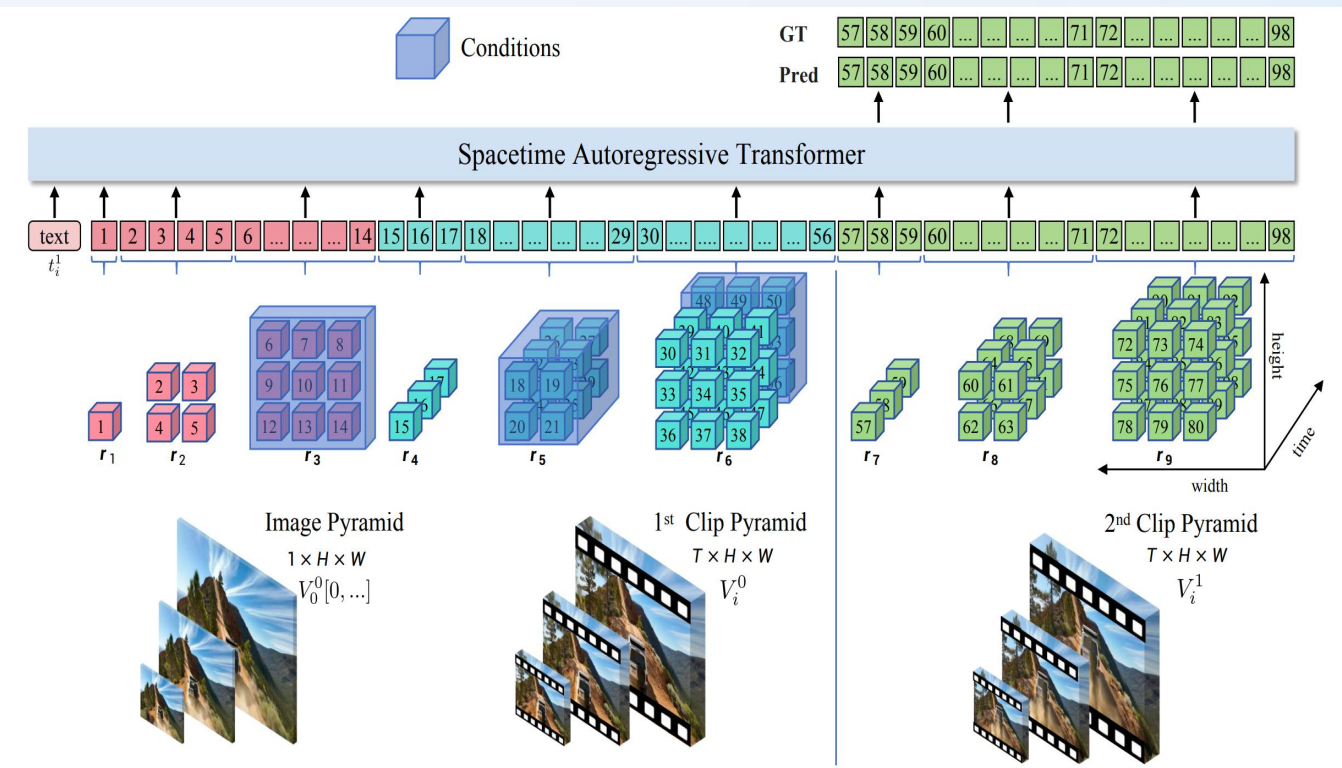
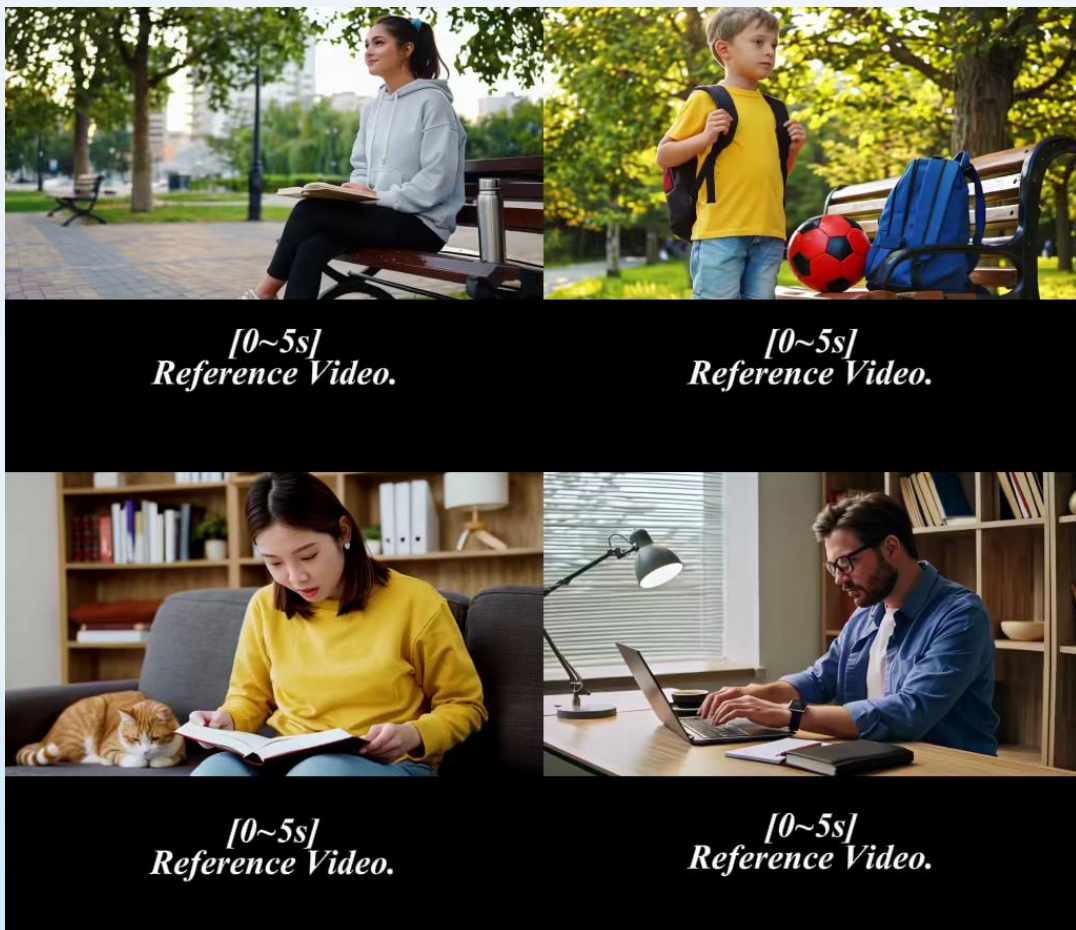


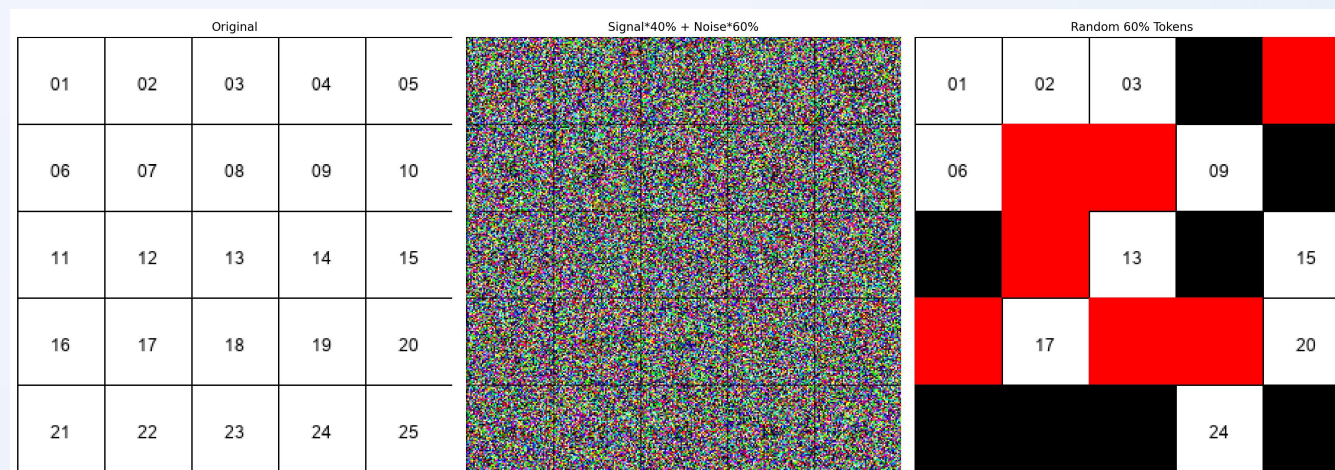
Figure 10: **Framework of InfinityStar-Interact.** We propose Semantic-Detail conditions (illustrated in light blue cubes) to control video synthesizing when interacting with users. It delivers superior visual and semantic consistency, as well as strong prompt-following capabilities.

04

生成式精炼网络：GRN Generative Refinement Networks

Limitations of current Diffusion AR models

- Allocate uniform computational effort to samples with different complexities
- Compared with explicit token-space reasoning, reasoning in a noisy latent space is less efficient



Credit to: AdaDiff: Adaptive Step Selection for Fast Diffusion Models

Hierarchical Binary Quantization

Algorithm 1 Hierarchical Binary Quantization

Require: VAE encoder $\mathcal{E}$, VAE decoder $\mathcal{D}$, an image or a video X , HBQ round M , $\delta(\cdot)$ function where $\delta(0) = -1$ and $\delta(1) = 1$

$Q_{queue} \leftarrow []$ ▷ HBQ binary labels

$F = \mathcal{E}(X)$ ▷ VAE Encoding

$F = \tanh(F)$ ▷ restrict data range to $[-1, 1]$

$c_1 = \text{ZerosLike}(F)$ ▷ initialize bucket centroids

for $i = 1, 2, \dots, M$ **do** ▷ HBQ rounds

$q_i = (F > c_i)$ ▷ binary quantization

 QueuePush(Q_{queue}, q_i) ▷ update HBQ binary labels

$c_{i+1} = c_i + \delta[q_i] \cdot 2^{-i}$ ▷ update bucket centroids for next round

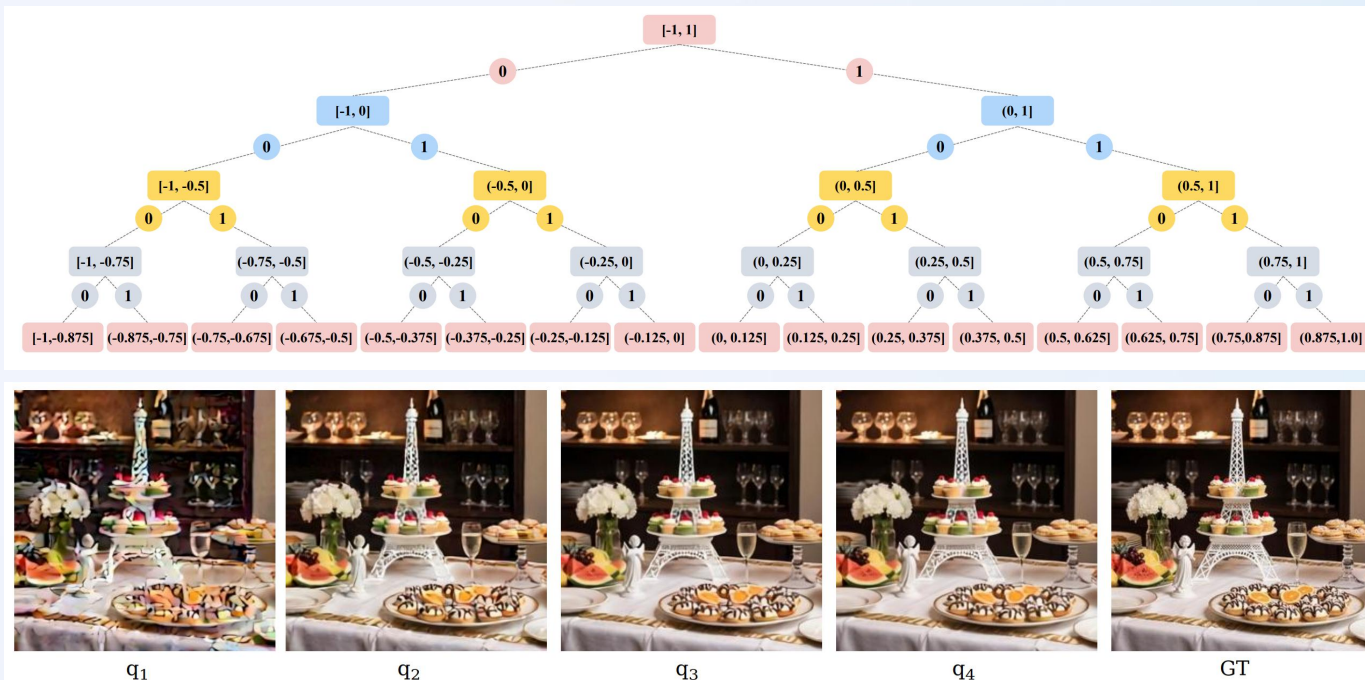
end for

$\hat{F} = \delta[q_1] \cdot 2^{-1} + \delta[q_2] \cdot 2^{-2} + \dots + \delta[q_M] \cdot 2^{-M}$ ▷ obtain quantized feature

$\hat{F} = \text{StopGrad}[\hat{F} - F] + F$ ▷ Straight-Through Estimator

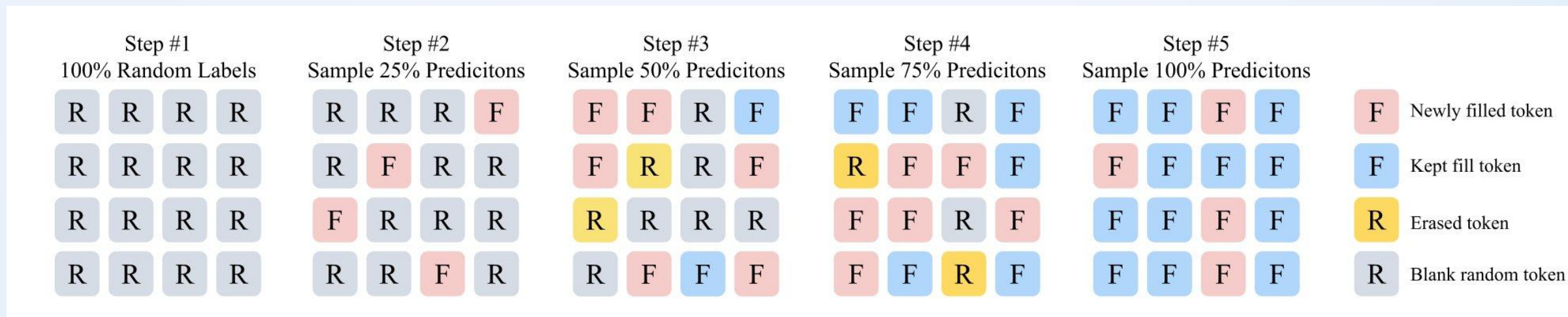
$\hat{X} = \mathcal{D}(\hat{F})$ ▷ VAE Decoding

Ensure: $Q_{queue} = [q_1, q_2, \dots, q_M], \hat{F}, \hat{X}$



- Our multi-round quantization process generates binary tokens sequentially, with each round corresponding to a bit
- The HBQ binary tokens are ordered from coarse to fine. Early tokens in the sequence correspond to core semantic concepts, while later tokens introduce high-frequency details

GRN Framework



- Starting from a random token map, GRN randomly selects more predictions at each step and refines all input tokens
- Compared to the second step, the third step filled six new tokens (pink), kept two tokens (blue), erased two tokens (yellow), and left six tokens blank (gray)

HBQ Image Tokenizer

Table 1: **Reconstruction performance comparison of image tokenizers on ImageNet (256×256)**

Method	Tokenizer Type	Spatial Ratio	Latent Channel	Channel Bits	Compress Ratio	rFID↓	LPIPS↓	SSIM↑	PSNR↑
SD-VAE [47]	Continuous	16	16	16	24	0.87	-	0.68	24.08
RAE [67]	Continuous	16	768	16	0.5	0.62	0.25	0.44	19.20
VAR [†] [52]	Discrete	16	N/A	N/A	193	0.85	0.15	0.64	22.47
LlamaGen [50]	Discrete	16	1	14	439	2.19	-	0.68	20.79
Open-MAGVIT2 [36]	Discrete	16	1	18	341	1.17	-	-	22.64
HBQ (M=4)	Discrete	16	16	4	96	0.56	0.13	0.71	23.01

- Our tokenizer demonstrates state-of-the-art reconstruction performance on the 256×256 ImageNet benchmark
- It achieves a remarkable rFID of 0.56. This result significantly surpasses not only the continuous SD-VAE (0.87 rFID) while operating at a $4\times$ higher compression rate

HBQ Joint Image and Video Tokenizer

Table 2: Reconstruction performance comparison of video tokenizers. To evaluate reconstruction quality, we curated a challenging validation set of 160 high-motion videos. For continuous tokenizers, we consider each latent channel to hold 16 bits of information. An HBQ tokenizer with M rounds introduces M bits within each latent channel.

Method	Tokenizer Type	Latent Channel	Spatial Stride	Temporal Stride	Channel Bits	Compress Ratio	rFVD↓	LPIPS↓	SSIM↑	PSNR↑
Wan 2.1 (patchify)	Continuous	64	16	4	16	24	19.5	0.058	0.929	34.10
Wan 2.2 (patchify)	Continuous	48	16	4	16	27	22.6	0.052	0.932	34.54
HBQ (w/o quant)	Continuous	16	16	4	16	96	144.6	0.141	0.879	31.14
HBQ (M=4)	Discrete	16	16	4	4	384	163.6	0.148	0.872	30.40
HBQ (M=6)	Discrete	16	16	4	6	256	148.8	0.142	0.878	30.98
HBQ (M=8)	Discrete	16	16	4	8	192	144.9	0.141	0.879	31.10
HBQ (w/o quant)	Continuous	64	16	4	16	24	43.2	0.078	0.935	34.79
HBQ (M=4)	Discrete	64	16	4	4	96	50.6	0.084	0.930	33.97
HBQ (M=4, tune λ_{GAN})	Discrete	64	16	4	4	96	30.1	0.078	0.928	33.98

- Reconstruction metrics such as rFVD and PSNR consistently improve as the number of HBQ rounds increases
- Expanding the latent channels from 16 to 64 boosts PSNR from 30.40 to an impressive 33.97. Notably, this performance is comparable to Wan 2.1 tokenizer, but is achieved at a 4× higher

GRN Class-to-Image Results



- The high-quality class-to-image visual samples generated by GRN-G with a gFID of 1.81

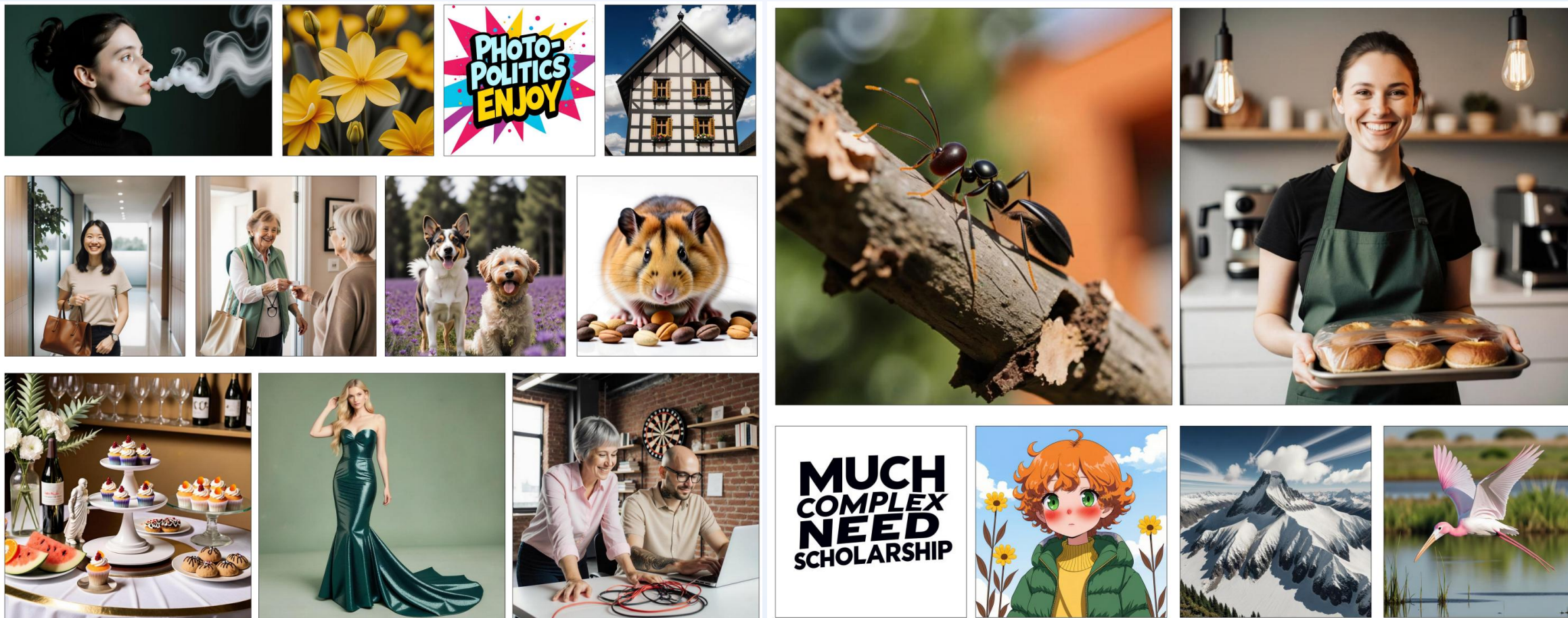
GRN Class-to-Image Results

Table 3: **Reference results on ImageNet 256×256 .** FID [27] and IS [48] of 50K samples are evaluated. Tokenizer “D” refers to a discrete tokenizer, while “C” refers to a continuous tokenizer.

Type	Model	Tokenizer	Loss	#Param	FID↓	IS↑
Diffusion	DiT-L/2 [43]	C	MSE	458M	5.02	167.2
Diffusion	DiT-XL/2 [43]	C	MSE	675M	2.27	278.2
Flow	SiT-XL/2 [37]	C	MSE	675M	2.06	277.5
Flow	REPA [62], SiT-XL/2	C	MSE	675M	1.42	305.7
Flow	RAE [67], DiT ^{DH} -XL/2	C	MSE	839M	1.13	262.6
Flow	JiT-B/16 [32]	C	MSE	131M	3.66	275.1
Flow	JiT-L/16 [32]	C	MSE	459M	2.36	298.5
Flow	JiT-H/16 [32]	C	MSE	953M	1.86	303.4
Flow	JiT-G/16 [32]	C	MSE	2B	1.82	292.6
Hybrid	MAR [33]	C	MSE	943M	1.55	303.7
Hybrid	BitDance-H-1x [2]	D	MSE	1B	1.24	304.4
AR	LlamaGen-L [50]	D	CE	343M	3.07	256.1
AR	LlamaGen-XL [50]	D	CE	775M	2.62	244.1
AR	LlamaGen-XXL [50]	D	CE	1.4B	2.34	253.9
AR	MaskGIT [7]	D	CE	227M	6.18	182.1
AR	VAR-d20 [52]	D	CE	600M	2.57	302.6
AR	VAR-d24 [52]	D	CE	1B	2.09	312.9
AR	VAR-d30 [52]	D	CE	2B	1.92	323.1
AR	RandAR-XXL [42]	D	CE	1.4B	2.15	322.0
AR	GRN-B	D	CE	130M	3.56	280.3
AR	GRN-L	D	CE	458M	2.64	314.8
AR	GRN-H	D	CE	952M	2.06	316.1
AR	GRN-G	D	CE	2B	1.81	299.0

- GRN is benchmarked against state-of-the-art diffusion, hybrid, and autoregressive models on the ImageNet 256×256 class-conditional generation task
- Our largest variant, GRN-G, achieves a state-of-the-art FID of 1.81, rivaling top diffusion and hybrid models.
- Notably, GRN-G surpasses foundational models like DiT and SiT in both FID and Inception Score

GRN Text-to-Image Results



GRN Text-to-Image Results

Table 4: Evaluation of Text-to-Image generation on GenEval [21].

Model	#Param	#Data	Single Obj.	Two Obj.	Count	Colors	Pos.	Color Attri.	Overall↑
<i>Proprietary Models</i>									
GPT Image 1 [40]	N/A	N/A	0.99	0.92	0.85	0.92	0.75	0.61	0.84
Seedream 3.0 [20]	N/A	N/A	0.99	0.96	0.91	0.93	0.47	0.80	0.84
<i>Diffusion Models</i>									
PixArt- α [9]	0.6B	N/A	0.98	0.50	0.44	0.80	0.08	0.07	0.48
SD3 Medium [18]	2B	N/A	0.98	0.74	0.63	0.67	0.34	0.36	0.62
JanusFlow [38]	1.3B	N/A	0.97	0.59	0.45	0.83	0.53	0.42	0.63
FLUX.1-Dev [31]	12B	N/A	0.98	0.81	0.74	0.79	0.22	0.45	0.66
SD3.5-Large [18]	8B	N/A	0.98	0.89	0.73	0.83	0.34	0.47	0.71
Lumina-Image-2.0 [45]	2.6B	111M	-	0.87	0.67	-	-	0.62	0.73
Show-o2 [57]	N/A	N/A	1.00	0.87	0.58	0.92	0.52	0.62	0.76
Z-Image-Turbo [5]	6B	N/A	1.00	0.95	0.77	0.89	0.65	0.68	0.82
HiDream-I1-Full [6]	17B	N/A	1.00	0.98	0.79	0.91	0.60	0.72	0.83
Z-Image [5]	6B	N/A	1.00	0.94	0.78	0.93	0.62	0.77	0.84
Qwen-Image [46]	20B	N/A	0.99	0.92	0.89	0.88	0.76	0.77	0.87
BAGEL [13]	14B	N/A	0.98	0.95	0.84	0.95	0.78	0.77	0.88
<i>Autoregressive Models</i>									
Emu3-Gen [56]	8B	N/A	0.98	0.71	0.34	0.81	0.17	0.21	0.54
Infinity [†] [26]	2B	160M	-	0.85	-	-	0.49	0.57	0.73
Janus-Pro [11]	N/A	N/A	0.99	0.89	0.59	0.90	0.79	0.66	0.80
Tar [25]	N/A	N/A	0.98	0.92	0.83	0.85	0.80	0.65	0.84
NextStep-1 [51]	14B	N/A	-	-	-	-	-	-	0.73
BitDance [3]	14B	N/A	1.00	0.96	0.71	0.95	0.72	0.83	0.86
GRN[†]	2B	80M	0.99	0.90	0.72	0.84	0.52	0.60	0.76

- GRN achieves an overall score of 0.76 on the GenEval benchmark
- When compared at an equivalent scale, GRN demonstrates superior performance
- GRN significantly outperforming models of a similar 2B size like SD3 Medium and Infinity

GRN Text-to-Video Results



GRN Text-to-Video Results

Table 5: Evaluation on the VBench benchmark. † result is with prompt rewriting.

Models	# Params	Human Action	Scene	Multiple Objects	Appear. Style	Quality Score	Semantic Score	Overall
Diffusion / Flow Models								
AnimateDiff-V2 [23]	1.5B	92.60	50.19	36.88	22.42	82.90	69.75	80.27
VideoCrafter-2.0 [8]	1.5B	95.00	55.29	40.66	25.13	82.20	73.42	80.44
OpenSora V1.2 [68]	1.1B	85.80	42.47	58.41	23.89	80.71	73.30	79.23
Show-1 [63]	6B	95.60	47.03	45.47	23.06	80.42	72.98	78.93
URSA [15]	1.7B	-	52.30	70.60	-	83.40	78.50	82.40
CogVideoX-5B [59]	5B	99.40	53.20	62.11	24.91	82.75	77.04	81.61
HunyuanVideo [29]	13B	94.40	53.88	68.55	19.80	85.09	75.82	83.24
Wan 2.1 [54]	14B	98.80	53.67	81.44	21.13	85.64	80.95	84.70
AutoRegressive Models								
Nova† [14]	0.6B	95.20	54.06	77.52	20.92	80.39	79.05	80.12
Emu3 [56]	8B	77.71	37.11	44.64	20.92	84.09	68.43	80.96
Lumos-1 [49]	3.6B	-	-	-	-	79.50	73.50	78.30
InfinityStar† [35]	8B	96.43	52.08	78.66	21.81	84.73	79.78	83.74
GRN†	2B	93.75	50.44	70.83	21.30	84.41	77.35	82.99

- GRN exhibits superior performance in generating videos from textual prompts
- When compared against diffusion and flowd models—including AnimateDiff2, VideoCraft2, OpenSora, Show-1, and CogVideoX, GRN achieves significantly higher scores across quality, semantic, and overall scores

05

核心要点

核心要点

➤ Infinity

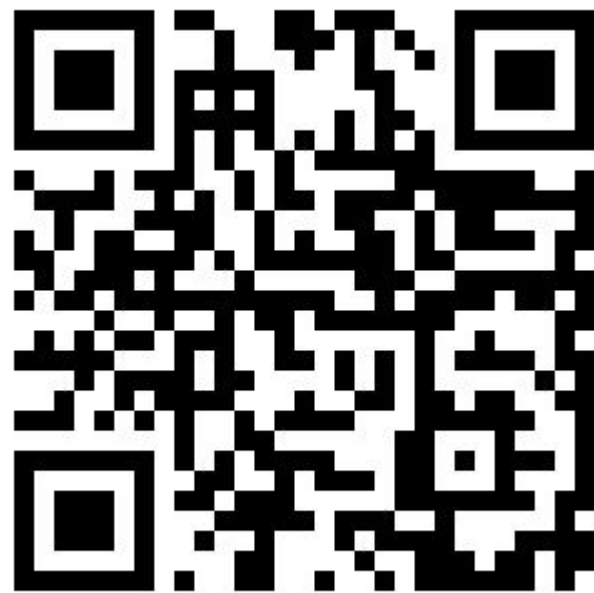
- bit 级建模提升了离散生成模型的表现
- 通过扩展词表与模型规模，实现接近连续建模的性能
- 强大的提示词遵循能力，并可生成高质量图像

➤ InfinityStar

- 统一且通用的视频生成模型
- 最先进的离散自回归文本到视频（T2V）模型
- 推理延迟实现 10 倍加速

➤ GRN: 既不是扩散，也不是自回归——视觉生成的第三路径

- 像艺术家一样进行全局式优化与细化
- 根据内容复杂度进行自适应生成
- 在图像与视频任务上都达到新的 SOTA



欢迎 GRN

自回归模型 在视觉生成领域中的应用

THANKS



AiCon

全球人工智能开发与应用大会

上海站

探索 AI 应用边界

Explore the limits of AI applications

2026 年 6 月 26-27 日 / 上海张江科学会堂



参会咨询

专题：世界模型与多模态智能突破

专题出品人：朱政

极佳视界 / 联合创始人 & 首席科学家



专题：Agent 系统架构与工程化实践

专题出品人：鲁浩楠

OPPO / 大模型算法负责人



专题：AI 原生数据工程

专题出品人：王彦辉

火山引擎 / 数智平台产品总监



专题：Agent 安全、评测与可信治理

专题出品人：马传雷（岳立）

支付宝 / 业务风险技术部负责人



专题：Agent 数据、记忆与运行时基础设施

专题出品人：邓亚峰

EverMind / CEO



专题：企业级研发体系的重构

专题出品人：沈浪

快手 / 研发效能负责人

